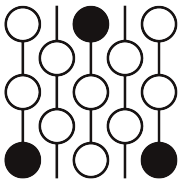
Review of Problem 6 – Siege

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: N/A

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- −1: blatant plagiarism or hallucinations

Score:-1

> **Reviewer 1**
>
> Blatant plagiarism in Section 7 of work already in the literature (found with AI-assistance). Proof could not be understood under the time constraint, but if all the Lemmas in Section 7 are correct we believe the overall proof can be saved. Exposition is also hard to follow and parse.

2 Submission with inline comments

Problem

Does a 3-manifold obtained by Dehn surgery on an L-space knot have at most two prime summands?

Result

Throughout, an *L-space knot* is a knot $K \subset S^3$ such that $S^3_r(K)$ is an L-space for some slope $r \in \mathbb{Q}$. The answer is **yes**, for every slope.

Main Theorem. *Let $K \subset S^3$ be an L-space knot and $r \in \mathbb{Q} \cup \{\infty\}$. Then $S^3_r(K)$ has at most two prime summands.*

The bound is sharp. By Moser [13], $S^3_6(T(2,3)) \cong L(2,1)\#L(3,2)$, which has two prime summands. More generally, if a cable $C_{a,b}(J)$ (with winding number a) is an L-space knot, then its surgery at the cabling slope ab is $L(a,b)\#S^3_{b/a}(J)$ [3].

Overview of the proof

Suppose $Y = S^3_r(K)$ has at least three prime summands. Classical results (Gabai; Gordon–Luecke) force $r = p$ to be a nonzero integer and K to be nontrivial. Since $\pi_1(Y)$ is normally generated by one element, Howie's solution of the Scott–Wiegold conjecture shows that at most two summands have nontrivial first homology; hence some summand B is an integer homology sphere other than S^3. A theorem of Gordon–Luecke supplies a lens space summand L of order $a \geq 2$.

Two independent arguments then collide. *Topologically*, K is fibred (Ni, Ghiggini), and in the $|p|$-fold cyclic cover of Y the summand B lifts to $|p|$ copies, while the fibred structure shows that the cover has fundamental group generated by $2g$ elements. Grushko's theorem then gives $|p| \leq g$. *Floer-theoretically*, the lens space summand forces $HF^+(S^3_p(K), i)$ to be invariant, up to grading shift, under $i \mapsto i + |p|/a$ in the labelling of Spin^c structures by $\mathbb{Z}/|p|$. Using the Ozsváth–Szabó integer surgery formula we compute, for $2 \leq |p| \leq 2g-2$, a translation-invariant multiset τ of each such group: its size depends only on how many integers $s \equiv i \pmod{|p|}$ lie in $[-(g-1), g-1]$, and its elements depend only on these integers. The size count forces $|p| \mid 2g-1$, and a symmetry argument comparing the class 0 (whose τ has repeated entries) with any other class in the same orbit (whose τ has distinct entries) rules out $|p| \mid 2g-1$, $|p| < 2g-1$. Since $|p| \leq g$, this is a contradiction.

The cover bound (Proposition 6.1), the computation of τ (Proposition 7.3) and the arguments of Sections 4–8 are argued here; the inputs we use from the literature are collected, with references, in Section 3.

3 Conventions and cited results

We write $\mathbb{F} = \mathbb{Z}/2$. For a knot $K \subset S^3$, M_K is its exterior, μ and λ are its meridian and Seifert longitude on $T = \partial M_K$, and $g = g(K)$ is its genus. Prime decompositions are those of Kneser and Milnor [9, 11]: every closed oriented 3-manifold $Y \neq S^3$ is a connected sum $P_1\# \cdots \# P_k$ of prime manifolds $P_i \neq S^3$, and k and the P_i are unique (up to order and homeomorphism); k is the number of prime summands, and S^3 has none. A closed manifold with at least two prime summands is reducible, since a splitting sphere does not bound a ball.

Mirroring. If mK denotes the mirror of K, then $S^3_r(mK) \cong -S^3_{-r}(K)$, and reversing orientation does not change the number of prime summands. An L-space surgery is never at slope 0 (it has $b_1 = 1$). If K has an L-space surgery at a negative slope, then mK has one at a positive slope. Hence we may assume that K is a *positive L-space knot*, meaning that it has an L-space surgery at some positive slope. The surgery slope p studied below is then still an arbitrary integer of either sign, and both signs are treated.

We use the following known results.

Cited Theorem A (Gabai [1])**.** *If $K \subset S^3$ is a nontrivial knot, then $S^3_0(K)$ is irreducible.*

Cited Theorem B (Gordon–Luecke [4])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then r is an integer.*

Cited Theorem C (Howie [8]; the Scott–Wiegold conjecture)**.** *A free product of three nontrivial cyclic groups is not the normal closure of a single element.*

Cited Theorem D (Gordon–Luecke [5])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then one of its prime summands is a lens space (other than S^3 and $S^1 \times S^2$).*

We use only the weak consequence that some prime summand is an L-space L with $|H_1(L)| \geq 2$ (lens spaces are L-spaces [18]).

Cited Theorem E (Ni [15]; Ghiggini [2] for $g = 1$; Ozsváth–Szabó [18])**.** *An L-space knot K is fibred, and its fibre is a minimal genus Seifert surface F, of genus g, with one boundary component.*

(Ozsváth and Szabó [18] showed that the knot Floer homology of an L-space knot in its top Alexander grading is $\mathbb{F}$; Ni and Ghiggini showed that this characterises fibred knots.)

Cited Theorem F (Grushko [6], Neumann [14]; see also [10])**.** *If G_1, G_2 are finitely generated groups, then* $\operatorname{rank}(G_1 * G_2) = \operatorname{rank} G_1 + \operatorname{rank} G_2$*, where* rank *is the minimal number of generators.*

Together with van Kampen's theorem, $\pi_1(M_1 \# M_2) \cong \pi_1(M_1) * \pi_1(M_2)$, this gives $\operatorname{rank} \pi_1(M_1 \# M_2) = \operatorname{rank} \pi_1(M_1) + \operatorname{rank} \pi_1(M_2)$.

Cited Theorem G (Perelman [20, 21, 22]; see [12])**.** *A closed simply connected 3-manifold is homeomorphic to S^3.*

This can be avoided; see Remark 6.5.

Heegaard Floer input. An *L-space* is a rational homology sphere Y with $HF^+(Y, \mathfrak{t}) \cong \mathcal{T}^+ := \mathbb{F}[U, U^{-1}]/U \cdot \mathbb{F}[U]$ for every $\mathfrak{t} \in \operatorname{Spin}^c(Y)$ [18]. Let K be a positive L-space knot of genus $g \geq 1$ and $p \neq 0$ an integer.

(H1) *The integers V_s* ([19, 18]; the notation is that of Ni–Wu [16]). There are integers V_s, $s \in \mathbb{Z}$, with

$$V_s \geq 0, \qquad V_s - V_{s+1} \in \{0, 1\}, \qquad V_s = 0 \iff s \geq g, \qquad V_{-s} = V_s + s.$$

Put $H_s := V_{-s}$; thus $H_s = V_s + s$, and $H_s = 0 \iff s \leq -g$.

(H2) *The integer surgery mapping cone, plus version* (Ozsváth–Szabó [19]). The set $\operatorname{Spin}^c(S^3_p(K))$ is labelled by $\mathbb{Z}/|p|$. For each label i, $HF^+(S^3_p(K), i)$ is isomorphic, as a relatively $\mathbb{Z}$-graded $\mathbb{F}[U]$-module, to the homology of the mapping cone of

$$D = \sum_s (v_s + h_s) \colon \bigoplus_{s \equiv i} A^+_s \longrightarrow \bigoplus_{s \equiv i} B^+_s$$

(direct sums over $s \equiv i \pmod{|p|}$), where $v_s \colon A_s^+ \to B_s^+$ and $h_s \colon A_s^+ \to B_{s+p}^+$ are homogeneous chain maps. For a positive L-space knot, $H_*(A_s^+) \cong H_*(B_s^+) \cong \mathcal{T}^+$, and on homology v_s is U^{V_s} and h_s is U^{H_s} [18, 19].

Why homology-level maps suffice. $\mathbb{F}[U]$ is a graded principal ideal domain, hence hereditary. So every complex of graded $\mathbb{F}[U]$-modules is quasi-isomorphic to its homology (it is formal), and in the derived category $\mathrm{Hom}(X, Z) = \mathrm{Hom}(H(X), H(Z)) \oplus \mathrm{Ext}^1(H(X), H(Z))$, with the appropriate degree shift. The target homology $\bigoplus \mathcal{T}^+$ is injective: $\mathcal{T}^+$ is divisible, $\mathbb{F}[U]$ is Noetherian, and over a Noetherian ring direct sums of injective modules are injective. So the Ext^1 term vanishes, and the morphism D is determined up to isomorphism in the derived category by its homology map. The mapping cone of a morphism is determined up to isomorphism by the morphism. Hence the homology of the cone, with its relative grading, can be computed from the homology-level maps U^{V_s} and U^{H_s}.

(H3) *Affine labelling* [19]. The labelling $\mathrm{Spin}^c(S_p^3(K)) \leftrightarrow \mathbb{Z}/|p|$ is affine over $H_1(S_p^3(K)) = \mathbb{Z}/|p| = \langle [\mu] \rangle$, and $[\mu]$ acts by $i \mapsto i + 1$. (The sign convention does not matter below.)

(H4) *Connected sums* (Ozsváth–Szabó [17]). $\mathrm{Spin}^c(Y_1 \# Y_2) = \mathrm{Spin}^c(Y_1) \times \mathrm{Spin}^c(Y_2)$, equivariantly for $H_1(Y_1) \oplus H_1(Y_2)$, and $CF^-(Y_1 \# Y_2, (\mathfrak{t}_1, \mathfrak{t}_2)) \simeq CF^-(Y_1, \mathfrak{t}_1) \otimes_{\mathbb{F}[U]} CF^-(Y_2, \mathfrak{t}_2)$. If L is an L-space then $CF^-(L, \mathfrak{t}) \simeq \mathbb{F}[U]$, so $HF^+(L \# Y'', (\mathfrak{t}, \mathfrak{t}'')) \cong HF^+(Y'', \mathfrak{t}'')$ as relatively graded $\mathbb{F}[U]$-modules, up to a shift. Lens spaces are L-spaces [18].

4 Reduction

Suppose $S_r^3(K)$ has at least three prime summands; we derive a contradiction.

If K is the unknot, its surgeries are S^3, $S^1 \times S^2$ and lens spaces, each with at most one prime summand. So K is nontrivial. Slope ∞ gives S^3, and slope 0 gives an irreducible manifold by Theorem A. By Theorem B, a non-integral slope also gives an irreducible manifold. An irreducible closed 3-manifold has at most one prime summand.

So $r = p$ is a nonzero integer. After mirroring (Section 3), K is a nontrivial positive L-space knot of genus $g \geq 1$, and $p \neq 0$ has either sign. Write

$$Y = S_p^3(K) = P_1 \# \cdots \# P_k, \qquad P_i \text{ prime}, \ k \geq 3.$$

$H_1(Y) = \mathbb{Z}/|p|$ is finite. So no P_i is $S^1 \times S^2$, and each $H_1(P_i)$ is a direct summand of $\mathbb{Z}/|p|$, hence finite cyclic.

5 Two topological consequences

Lemma 5.1 (a homology sphere summand)**.** *Some P_i is an integer homology sphere $B \neq S^3$.*

Proof. $\pi_1(M_K)$ is normally generated by μ, and $\pi_1(Y)$ is a quotient of it, so $\pi_1(Y)$ is normally generated by one element. Suppose three summands, say P_1, P_2, P_3, had nontrivial H_1. Then $\pi_1(Y) = \pi_1(P_1) * \cdots * \pi_1(P_k)$ maps onto $\pi_1(P_1) * \pi_1(P_2) * \pi_1(P_3)$, by killing the other factors, and then onto $H_1(P_1) * H_1(P_2) * H_1(P_3)$. This target is a free product of three nontrivial cyclic groups, and it is normally generated by the image of one element. This contradicts Theorem C. So at most two summands have nontrivial H_1. Since $k \geq 3$, some P_i has $H_1(P_i) = 0$, and $P_i \neq S^3$ because it is a prime summand. □

Write $Y \cong A\#B$, where B is this homology sphere and A is the connected sum of the other summands.

Lemma 5.2 (an L-space summand)**.** *$Y \cong L\#Y''$, where L is a lens space with $a := |H_1(L)| \geq 2$. Hence a divides $|p|$ and $|p| \geq 2$.*

Proof. Y is reducible since $k \geq 2$. By Theorem D, some prime summand is a lens space $L \neq S^3, S^1 \times S^2$, so $a \geq 2$. Also $H_1(L)$ is a direct summand of $\mathbb{Z}/|p|$. □

6 The cover bound

Proposition 6.1 (cover bound)**.** *Let $K \subset S^3$ be fibred, with fibre F of genus g and one boundary component, and let $p \neq 0$ be an integer. If $S^3_p(K) \cong A\#B$ with B an integer homology sphere other than S^3, then, with $N := |p|$ and $\tilde{A}$ the connected N-fold cyclic cover of A defined below,*

$$|p| \cdot \operatorname{rank} \pi_1(B) + \operatorname{rank} \pi_1(\tilde{A}) \leq 2g, \qquad \text{and hence} \qquad |p| \leq g.$$

Setup. $Y = S^3_p(K) = M_K \cup_T V$, where the meridian disc of the solid torus V is bounded by $\alpha = p\mu + \lambda$ on $T = \partial M_K$. $H_1(M_K) = \mathbb{Z}\langle\mu\rangle$; the class of a loop is its algebraic intersection number with F, and $\lambda = 0$ in $H_1(M_K)$. So $H_1(Y) = \mathbb{Z}/N$. Let $\varphi\colon \pi_1(Y) \to H_1(Y) = \mathbb{Z}/N$ be the abelianisation, which is onto, and let $q\colon Y_N \to Y$ be the connected N-fold cyclic cover with group $\ker \varphi$.

Lemma 6.2 (lower bound)**.** *$Y_N \cong \tilde{A}\#B\#\cdots\#B$ with N copies of B, and* $\operatorname{rank} \pi_1(Y_N) = \operatorname{rank} \pi_1(\tilde{A}) + N \operatorname{rank} \pi_1(B)$ *with* $\operatorname{rank} \pi_1(B) \geq 2$.

Proof. 1. Let S be the splitting sphere, so $Y = A^\circ \cup_S B^\circ$, where $^\circ$ means that one open ball is removed.

2. $\pi_1(Y) = \pi_1(A) * \pi_1(B)$, and $H_1(A) = H_1(Y) \cong \mathbb{Z}/N$ because $H_1(B) = 0$.

3. The restriction of φ to $\pi_1(B^\circ) = \pi_1(B)$ factors through $H_1(B) = 0$. So $q^{-1}(B^\circ)$ is N disjoint copies of B°, each mapped homeomorphically onto B°.

4. S is simply connected, so $q^{-1}(S)$ is N spheres.

5. The restriction of φ to $\pi_1(A^\circ) = \pi_1(A)$ is the surjection $\pi_1(A) \to H_1(A) \cong \mathbb{Z}/N$. So $q^{-1}(A^\circ)$ is connected: it is the induced N-fold cyclic cover $\tilde{A}$ of A with N balls removed (the removed ball lifts to N balls).

6. Therefore Y_N is $\tilde{A}$ minus N balls, with a copy of B° glued to each boundary sphere. The gluing graph is a star, hence a tree, so $Y_N \cong \tilde{A}\#B\#\cdots\#B$ with N copies of B. (If $N = 1$ the cover is trivial and $\tilde{A} = A$.)

7. By van Kampen and Theorem F, $\operatorname{rank} \pi_1(Y_N) = \operatorname{rank} \pi_1(\tilde{A}) + N \operatorname{rank} \pi_1(B)$.

8. $\pi_1(B)$ is perfect, since $H_1(B) = 0$, and nontrivial, by Theorem G, since $B \neq S^3$. A nontrivial cyclic group has nontrivial abelianisation, so $\pi_1(B)$ is not cyclic, and $\operatorname{rank} \pi_1(B) \geq 2$. □

Lemma 6.3 (upper bound)**.** $\operatorname{rank} \pi_1(Y_N) \leq 2g$.

Proof. 1. *The exterior lifts to a mapping torus.* On $\pi_1(M_K)$, φ is the composite $\pi_1(M_K) \to H_1(M_K) = \mathbb{Z} \to \mathbb{Z}/N$ with $\mu \mapsto 1$. So $E := q^{-1}(M_K)$ is connected and is the N-fold cyclic cover dual to F. It is the mapping torus of h^N, where h is the monodromy, and it fibres over S^1 with fibre $\tilde{F}$, a homeomorphic lift of F. So $\pi_1(E) = \pi_1(\tilde{F}) \rtimes \langle t \rangle$, and the fibration map $\pi\colon \pi_1(E) \to \mathbb{Z}$ has kernel $\pi_1(\tilde{F})$ and sends $t \mapsto 1$.

2. *The boundary lifts to one torus.* $\mu \in \pi_1(T)$ maps onto a generator of $\mathbb{Z}/N$. So $\tilde{T} := q^{-1}(T)$ is one torus, and $q|\colon \tilde{T} \to T$ has degree N.

3. *The solid torus lifts to one solid torus.* μ meets α once, so μ is isotopic in V to the core of V. Hence $\pi_1(V)$ maps onto $\mathbb{Z}/N$, and $\tilde{V} := q^{-1}(V)$ is one solid torus with $\partial\tilde{V} = \tilde{T}$.

4. *The filling curve.* The meridian disc of V lifts to N discs. Let $\beta \subset \tilde{T}$ be the boundary of one of them. Then q maps β homeomorphically onto α, so $q_*[\beta] = [\alpha]$, and $Y_N = E \cup \tilde{V}$ is the Dehn filling of E along β.

5. *Intersection numbers.* For a degree-N covering $f\colon \tilde{T} \to T$ of tori, $f_*x \cdot f_*y = N(x \cdot y)$ for $x, y \in H_1(\tilde{T})$. This follows from the transfer $f^!$: $f_*x \cdot z = x \cdot f^!z$ and $f^!f_* = N$, since deck transformations of a torus act trivially on H_1. Now $\tilde{F} \cap \tilde{T} = \partial\tilde{F}$ is a single curve $\tilde{\lambda}$ with $q_*[\tilde{\lambda}] = [\lambda]$. Hence
$$N(\beta \cdot \tilde{\lambda}) = \alpha \cdot \lambda = (p\mu + \lambda) \cdot \lambda = \pm p,$$
so $\beta \cdot \tilde{\lambda} = \pm 1$. This holds for either sign of p.

6. *β goes once around.* On $\pi_1(\tilde{T})$, the map π is algebraic intersection with the fibre boundary $\tilde{\lambda}$. So $\pi(\beta) = \pm 1$, and β represents $t^{\pm 1}w$ for some $w \in \pi_1(\tilde{F})$, for any choice of basepoint path.

7. *Conclusion.* By van Kampen, $\pi_1(Y_N) = \pi_1(E)/\langle\langle\beta\rangle\rangle$. In this quotient $t = w^{\mp 1}$ lies in the image of $\pi_1(\tilde{F})$. So $\pi_1(Y_N)$ is generated by the image of $\pi_1(\tilde{F})$, which is free of rank $2g$, and $\operatorname{rank}\pi_1(Y_N) \le 2g$. □

Proof of Proposition 6.1. Combining Lemmas 6.2 and 6.3: $2N \le N \operatorname{rank}\pi_1(B) \le N \operatorname{rank}\pi_1(B) + \operatorname{rank}\pi_1(\tilde{A}) = \operatorname{rank}\pi_1(Y_N) \le 2g$, so $|p| = N \le g$. □

Remark 6.4. The bound is attained: for $T(2,3)$ and $p = 1$, Y is the Poincaré sphere and $2 = 2$. For a nontrivial fibred knot J and $p \ge 2$, the cable $C_{p,1}(J)$ is fibred of genus $p\,g(J) \ge p$, and its p-surgery is $L(p,1)\#S^3_{1/p}(J)$ [3], in agreement with the bound. (This cable is not an L-space knot; Proposition 6.1 needs only fibredness.)

Remark 6.5 (avoiding Theorem G)**.** Replace rank by Heegaard genus h. Since $\beta \cdot \tilde{\lambda} = \pm 1$, Y_N is the closed manifold of an open book with page $\tilde{F}$, so $h(Y_N) \le 2g$. By Haken's lemma [7], $h(Y_N) = h(\tilde{A}) + Nh(B)$. Finally $h(B) \ge 2$, since the closed 3-manifolds of Heegaard genus at most 1 are S^3, $S^1 \times S^2$ and lens spaces. This again gives $N \le g$.

7 Heegaard Floer input

In this section K is a nontrivial positive L-space knot of genus g, and $p \ne 0$ is an integer with $N := |p|$.

7.1 Coset invariance

Suppose $S^3_p(K) \cong L\#Y''$, where L is an L-space with $a = |H_1(L)| \geq 2$ and Y'' is a closed oriented 3-manifold. If the homeomorphism reverses orientation, replace (L, Y'') by $(-L, -Y'')$; $-L$ is again an L-space of the same order. Put $b := N/a$. Then b is an integer and $b \not\equiv 0 \pmod N$.

Lemma 7.1 (coset invariance)**.** *For every label i, $HF^+(S^3_p(K), i) \cong HF^+(S^3_p(K), i+b)$ as relatively graded $\mathbb{F}[U]$-modules, up to a degree shift.*

Proof. $H_1(L)$ is a subgroup of order a of $\mathbb{Z}/N = \langle[\mu]\rangle$, so it is $\langle b[\mu]\rangle$. By (H4), the Spin^c structures with a fixed Y''-component $\mathfrak{t}''$ form one $H_1(L)$-orbit, and all of them have $HF^+ \cong HF^+(Y'', \mathfrak{t}'')$ up to shift. By (H3), the $H_1(L)$-orbits are exactly the cosets $i + b\mathbb{Z}$ of labels. □

7.2 The invariant τ

Definition 7.2. Let M be a graded $\mathbb{F}[U]$-module that is finite-dimensional in each degree. Record degrees by *level*, meaning half the grading, so that U lowers the level by 1. Define $\tau(M)$ to be the multiset of levels k, where k appears with multiplicity $\dim(M_k/U \cdot M_{k+1})$.

τ is an isomorphism invariant, and shifting the grading translates τ. If $M = \mathcal{T}^+ \oplus \bigoplus_l \mathbb{F}[U]/U^{e_l}$, then $\tau(M)$ is the multiset of the top levels of the finite summands, because U maps onto $\mathcal{T}^+$; in particular $|\tau(M)|$ is the number of finite summands. Clearly $\tau(M \oplus M') = \tau(M) \cup \tau(M')$.

7.3 The level model and the structure of each class

Fix a class $c \in \mathbb{Z}/N$. Let $s_1 < s_2 < \cdots < s_m$ be the integers $s \equiv c \pmod N$ with $|s| \leq g - 1$. Consecutive ones differ by N, and

$$m = n_c := \#\{s \equiv c \pmod N : \ |s| \leq g - 1\}.$$

Level model. By (H2) we may compute with the homology-level maps U^{V_s} and U^{H_s} between copies of $\mathcal{T}^+$.

- Record each tower $\mathcal{T}^+$ by its *bottom level*.
- Regrade each B-tower so that the cone maps preserve levels. This is possible: in one class the A's and B's form a connected graph (each B is joined to its at most two adjacent A's), and the relative grading of the cone is homogeneous. A map U^e from a tower with bottom x then lands in a tower with bottom $x + e$.
- Under this regrading, an A-element at level k maps to the B-element at level k in each adjacent B that is *present at level k*, meaning that its bottom is $\leq k$. All exponents are ≥ 0, so every present B has its adjacent A's present.
- So at level k the cone map is the coboundary $\delta_k \colon \mathbb{F}^{\text{vertices}} \to \mathbb{F}^{\text{edges}}$ of a graph G_k. Its vertices are the A's with bottom $\leq k$, and its edges are the B's with bottom $\leq k$, each B joining its adjacent A's.
- U acts on cochains by restriction from G_{k+1} to G_k.
- The cone has two terms and D commutes with U, so $H(\text{cone}) = \ker D \oplus \operatorname{coker} D$ as graded $\mathbb{F}[U]$-modules.

The true relative grading is 2·level+constant on each side, because both satisfy the same constraints on a connected graph. So τ, computed in levels, is the invariant τ above up to translation.

Proposition 7.3 (structure of a class)**.** *Let c and $s_1 < \cdots < s_m$ be as above.*

(A) Let $0 < p \le 2g-2$ and $m \ge 1$. Then $\tau(HF^+(S^3_p(K), c)) = \{y - 1 : \ y \in Y_c\}$, where Y_c is the multiset of the $m-1$ interior peak *levels*

$$y_j = \frac{\mu_j^2 - p^2/4}{2p}, \qquad \mu_j = s_j + \frac{p}{2} \qquad (1 \le j \le m-1).$$

In particular $|\tau| = m-1$.

(B) Let $p = -N < 0$ and $m \ge 1$. Let Π_c be the multiset of the $m+1$ peak *levels*

$$b(t) = -\frac{u^2 - N^2/4}{2N}, \qquad u = t + \frac{N}{2}, \qquad t \in \{s_1 - N,\, s_1,\, s_2, \ldots, s_m\},$$

and let M be its maximum. Then $\tau(HF^+(S^3_p(K), c))$ is $\{y - 1 : \ y \in \Pi_c\}$ with one copy of $M-1$ removed. In particular $|\tau| = m$.

Neither multiset depends on K.

Proof of (A). In class c the A's and B's form one infinite path $\cdots A_{s-p} - B_s - A_s - B_{s+p} - \cdots$. Writing $x(s)$ for the bottom of A_s, the bottoms must satisfy $x(s+p) - x(s) = H_s - V_{s+p}$, since B_{s+p} receives U^{H_s} from A_s and $U^{V_{s+p}}$ from A_{s+p}. One solution is

$$x(s) = \frac{s(s-p)}{2p} - V_s;$$

indeed $x(s+p) - x(s) = s - V_{s+p} + V_s = H_s - V_{s+p}$, using $H_s = V_s + s$. The B between s and $s+p$ has bottom

$$y = x(s) + H_s = \frac{s(s+p)}{2p} = \frac{\mu^2 - p^2/4}{2p}, \qquad \mu = s + \frac{p}{2}.$$

We have $x(s) \to +\infty$ as $|s| \to \infty$: for $s \ge g$, $V_s = 0$, and for $s < 0$, $V_s = V_{-s} + |s| \le g + |s|$. So the quadratic term dominates, and each G_k is a finite subforest of a path.

The coboundary of a finite forest is onto. So D is surjective and $HF^+ \cong \ker D$. At level k, $\ker \delta_k$ is the space of functions on the components of G_k, and U is restriction of functions. Hence $\dim(M_k / U \cdot M_{k+1})$ is the number of *merges*: the number of components of G_k minus the number of components of G_{k+1} that contain a vertex of G_k.

Edges with a zero exponent. $V_s = 0$ iff $s \ge g$, and $H_s = 0$ iff $s \le -g$. A B with both exponents 0 would be B_{s+p} with $s \le -g$ and $s + p \ge g$, so $p \ge 2g$, which is excluded. A B with exactly one exponent 0 appears at the same level as its endpoint A_e on that side. That endpoint is new at that level. Its other edge has a positive exponent at A_e, since $V_e = H_e = 0$ would need $e \ge g$ and $e \le -g$; so that other edge appears strictly later. The far endpoint of B is old, since B's exponent there is positive. So at this level B attaches the new vertex A_e to one old component, and merges no old components.

Edges with both exponents ≥ 1. The edge B_{s+p} between A_s and A_{s+p} has exponents $H_s \ge 1$ (iff $s \ge -(g-1)$) and $V_{s+p} \ge 1$ (iff $s + p \le g - 1$). So these are exactly the $m-1$ edges between consecutive s_j, s_{j+1}: the interior peaks. Each appears strictly after both of its endpoints, and since G is a path, each one joins two distinct components of the old graph.

So the merges from level k to level $k+1$ are the interior peaks at level $k+1$. Hence $\tau = \{y - 1 :$ y an interior peak$\}$ and $|\tau| = m - 1$. □

Proof of (B). Here $h_s\colon A_s \to B_{s-N}$. Let C be the class-c cone.

Truncation, step 1. Let $P = \{A_s : s < g\} \cup \{B_t : t < g\}$. It is a subcomplex: for $s < g$, A_s maps to B_s and B_{s-N}, and both indices are $< g$. The quotient $Q = C/P = \{A_s, B_s : s \geq g\}$ has differential $D_Q = v_s + h_s$, where $v_s\colon A_s \to B_s$ is the identity since $V_s = 0$, and h_s lowers the index by N and is present only while $s - N \geq g$. On finitely supported chains, the identity plus a part that lowers the index, with indices bounded below, is bijective: invert it by a finite Neumann series. So Q is acyclic and $H(C) \cong H(P)$.

Truncation, step 2. Let $P' = \{A_s : |s| < g\} \cup \{B_t : -g - N < t < g\}$. It is a subcomplex of P: for $|s| < g$ we have $s - N > -g - N$. The quotient $R_0 = P/P' = \{A_s : s \leq -g\} \cup \{B_t : t \leq -g - N\}$ has differential $h_s + v_s$, where $h_s\colon A_s \to B_{s-N}$ is the identity since $H_s = 0$, and $v_s\colon A_s \to B_s$ is present only while $s \leq -g - N$. Relative to the pairing $A_s \leftrightarrow B_{s-N}$, the second part raises the B-index by N, and indices are bounded above. So it is again identity plus nilpotent on finitely supported chains, and R_0 is acyclic. Hence $H(C) \cong H(P')$.

Shape of P'. The class-c integers in $(-g-N, g)$ are $s_1 - N, s_1, \ldots, s_m$, because $s_1 - N \leq -g < s_1$ and $s_m < g \leq s_m + N$. So P' is the zigzag

$$B_{s_1-N} - A_{s_1} - B_{s_1} - A_{s_2} - \cdots - A_{s_m} - B_{s_m}.$$

Every exponent on it is ≥ 1: $V_{s_j} \geq 1$ since $s_j \leq g - 1$, and $H_{s_j} = V_{-s_j} \geq 1$ since $-s_j \leq g - 1$. The two end B's each have one A-neighbour. Treat them as edges to a *ground* vertex, at which cochains take the value 0. Then $G \cup \{\text{ground}\}$ is a single cycle with $m + 1$ edges.

Levels. Put $x(s) = -\frac{s(s+N)}{2N} - V_s$ and $b(t) = -\frac{t(t+N)}{2N} = -\frac{u^2 - N^2/4}{2N}$, $u = t + \frac{N}{2}$. Then B_s has bottom $x(s) + V_s = b(s)$, and since $H_s = V_s + s$,

$$x(s) + H_s = b(s) + s = \frac{-s^2 - sN + 2sN}{2N} = -\frac{s(s-N)}{2N} = b(s-N),$$

as required for B_{s-N}.

The cokernel part. $\operatorname{coker} \delta_k = H^1(G_k \cup \text{ground}, \text{ground})$, which is $H^1(G_k \cup \text{ground})$ because the ground is one point. This is 0 until the cycle closes, at the maximal peak level M. From then on it is $\mathbb{F}$, and U (restriction of cochains) is an isomorphism. So the cokernel part is a single tower and contributes nothing to τ.

The kernel part. $\ker \delta_k$ is the space of functions on the components of G_k that do not contain the ground. A vertex that is new at level $k + 1$ has its edges strictly higher, since all exponents are ≥ 1. So it is isolated at level $k + 1$, and the new edges at level $k + 1$ join only old vertices or the ground. Hence $\dim(\ker_k / U \cdot \ker_{k+1})$ is the number of merges among old components, including merges into the ground component. That number is $\#(\text{new edges at level } k+1) - \Delta b_1$, and b_1 increases (by 1) only at level M. So $\tau = \{y - 1 : y \in \Pi_c\}$ with one copy of $M - 1$ removed, and $|\tau| = (m + 1) - 1 = m$. □

7.4 Divisibility

Proposition 7.4. *Suppose $2 \leq |p| \leq 2g - 2$ and $S^3_p(K) \cong L \# Y''$ with L an L-space of order $a \geq 2$. Then $|p|$ divides $2g - 1$.*

Proof. Write $2g - 1 = QN + r$ with $0 \leq r < N$. Then $Q \geq 1$, since $N \leq 2g - 2$. The integers in $[-(g-1), g-1]$ are $2g - 1$ consecutive integers, so $n_i = Q + 1$ on a set S of r cyclically consecutive residues, and $n_i = Q$ elsewhere. In particular every $n_i \geq 1$.

By Proposition 7.3, $|\tau(HF^+(S^3_p(K), i))|$ equals $n_i - 1$ if $p > 0$ and n_i if $p < 0$; either way it is a strictly increasing function of n_i. By Lemma 7.1 and the invariance of τ up to translation, $|\tau|$ is invariant under $i \mapsto i + b$ with $b \not\equiv 0$. So S is invariant under translation by b.

If $0 < r < N$, then S is a proper nonempty cyclic interval in $\mathbb{Z}/N$. It has exactly one start point (an $x \in S$ with $x - 1 \notin S$), and a translation preserving S must fix that start point. That is impossible for $b \not\equiv 0$. So $r = 0$, i.e. N divides $2g - 1$. □

7.5 The key contradiction

Theorem 7.5. *Let K be a nontrivial positive L-space knot of genus g, and p a nonzero integer with $|p| \geq 2$, $|p|$ dividing $2g - 1$ and $|p| < 2g - 1$. Then $S^3_p(K)$ is not homeomorphic to $L\#Y''$ for any L-space L with $|H_1(L)| \geq 2$ and any closed oriented 3-manifold Y''.*

Proof. Let $N := |p|$ and $n := (2g - 1)/N$. Both are odd, $N \geq 3$ and $n \geq 3$. In particular $|p| \leq (2g - 1)/3 \leq 2g - 2$, so Proposition 7.3 applies. Suppose such a splitting exists, and let b be as in Lemma 7.1.

(i) The comparison. Let t be the representative of b with $0 < |t| \leq (N - 1)/2$; it exists because N is odd and $b \not\equiv 0$. By Lemma 7.1 and the invariance of τ:

$$(\star) \qquad \tau \text{ of class } 0 \text{ and } \tau \text{ of class } t \text{ agree up to a translation.}$$

(ii) Arithmetic. For $|c| \leq (N - 1)/2$, the integers $s \equiv c$ with $|s| \leq g - 1$ are exactly $c + kN$ for $|k| \leq (n - 1)/2$. Indeed $|c + kN| \leq (N - 1)/2 + (n - 1)N/2 = g - 1$, while $|k| \geq (n + 1)/2$ gives $|c + kN| \geq (nN + 1)/2 > g - 1$. So $m = n$ in every class.

Key fact. Let $u = c + (k + \frac{1}{2})N$ and $u' = c + (k' + \frac{1}{2})N$ with $k \neq k'$. If $u^2 = u'^2$, then $u = -u'$, so $2c + (k + k' + 1)N = 0$. Hence $N \mid 2c$, so $N \mid c$ because N is odd, and $c = 0$. So for $c = t$, distinct k give distinct values of u^2, hence distinct levels in both formulas of Proposition 7.3.

(iii) Case $p > 0$. Class 0: $s_j = (j - \frac{n+1}{2})N$, so the interior peaks have $\mu_j = s_j + N/2 = (j - \frac{n}{2})N$ for $1 \leq j \leq n - 1$. These are $\pm N/2, \pm 3N/2, \ldots, \pm(n - 2)N/2$: nonzero $\pm$ pairs. So every element of τ has even multiplicity, and τ is nonempty since $n - 1 \geq 2$. *Class t:* $\mu_j = t + (j - \frac{n+1}{2} + \frac{1}{2})N$, and by (ii) the $n - 1$ levels are distinct. A translate of a nonempty multiset in which every multiplicity is even never has all multiplicities equal to 1. This contradicts $(\star)$.

(iv) Case $p < 0$. Class 0: the peaks are $t' = kN$ for $k = -\frac{n+1}{2}, \ldots, \frac{n-1}{2}$, so $u = t' + N/2$ runs over $\pm N/2, \pm 3N/2, \ldots, \pm nN/2$, each value of $|u|$ exactly twice. The maximal level M comes from $|u| = N/2$. Removing one copy of $M - 1$ leaves the levels for $|u| = 3N/2, \ldots, nN/2$ still doubled, and these exist since $n \geq 3$. So τ has a repeated element. *Class t:* by (ii) the $n + 1$ peak levels are distinct, so the n elements of τ are distinct. This contradicts $(\star)$. □

8 Proof of the Main Theorem

Assume, as in Section 4, that K is a nontrivial positive L-space knot of genus g, $p \neq 0$, and $Y = S^3_p(K)$ has $k \geq 3$ prime summands.

1. By Lemma 5.1, $Y \cong A\#B$ with B an integer homology sphere, $B \neq S^3$.
2. By Lemma 5.2, $Y \cong L\#Y''$ with L a lens space, hence an L-space, of order $a \geq 2$. So $|p| \geq 2$.
3. K is fibred with fibre of genus g and one boundary component (Theorem E). Proposition 6.1 gives $|p| \leq g$. If $g = 1$, this contradicts $|p| \geq 2$. So $g \geq 2$, and $2 \leq |p| \leq g \leq 2g - 2$.
4. By Proposition 7.4, applied to the splitting of step 2, $|p|$ divides $2g - 1$. Also $|p| \leq g < 2g - 1$.
5. By Theorem 7.5, the splitting of step 2 cannot exist. This is a contradiction.

So $S^3_r(K)$ has at most two prime summands, for every L-space knot K and every slope r. □

9 Remarks

- For a fibred knot, a reducible integral surgery $S^3_p(K)$ with an integer homology sphere summand $B \neq S^3$ requires $|p| \leq g$ (Proposition 6.1). This uses only fibredness.
- Granting Theorem D, the argument of Section 8 shows more: a *reducible* surgery on an L-space knot never has a prime summand that is an integer homology sphere other than S^3, since steps 2–5 use only the homology sphere summand and the lens space summand. (Irreducible surgeries can be homology spheres, e.g. $S^3_{+1}(T(2,3))$ is the Poincaré sphere.) This agrees with the known examples, cables at the cabling slope, whose summands are $L(a,b)$ and $S^3_{b/a}(J)$.

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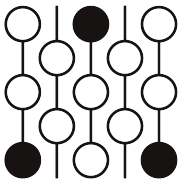
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Review of Problem 6 – Siege

First Proof Third Batch
Reviewer 2

October 8–9, 2026



1 Review

I was unable to determine in the time allotted whether this submission contains a correct solution to the problem. Through Section 6, the arguments appear to be essentially correct, though poorly written. Section 7 contains a long and technical proof of a Heegaard-Floer lemma that I was unable to evaluate; with LLM assistance, it was discovered that this section appears to be largely plagiarized from Hom-Lidman-Zufelt "Reducible surgeries and Heegaard Floer homology" which was cited by many of the other submissions. If the plagiarized result is correctly proven, then the claimed results follow.

The writing is easily the worst of the five submissions. The references/attribution appear to be complete and correct for the results used apart from the plagiarism in Section 7.

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: N/A

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: -1

2 Submission with inline comments

Problem

Does a 3-manifold obtained by Dehn surgery on an L-space knot have at most two prime summands?

Result

Throughout, an *L-space knot* is a knot $K \subset S^3$ such that $S^3_r(K)$ is an L-space for some slope $r \in \mathbb{Q}$. The answer is **yes**, for every slope.

Main Theorem. *Let $K \subset S^3$ be an L-space knot and $r \in \mathbb{Q} \cup \{\infty\}$. Then $S^3_r(K)$ has at most two prime summands.*

The bound is sharp. By Moser [13], $S^3_6(T(2,3)) \cong L(2,1)\#L(3,2)$, which has two prime summands. More generally, if a cable $C_{a,b}(J)$ (with winding number a) is an L-space knot, then its surgery at the cabling slope ab is $L(a,b)\#S^3_{b/a}(J)$ [3].

Overview of the proof

Suppose $Y = S^3_r(K)$ has at least three prime summands. Classical results (Gabai; Gordon–Luecke) force $r = p$ to be a nonzero integer and K to be nontrivial. Since $\pi_1(Y)$ is normally generated by one element, Howie's solution of the Scott–Wiegold conjecture shows that at most two summands have nontrivial first homology; hence some summand B is an integer homology sphere other than S^3. A theorem of Gordon–Luecke supplies a lens space summand L of order $a \geq 2$.

Two independent arguments then collide. *Topologically*, K is fibred (Ni, Ghiggini), and in the $|p|$-fold cyclic cover of Y the summand B lifts to $|p|$ copies, while the fibred structure shows that the cover has fundamental group generated by $2g$ elements. Grushko's theorem then gives $|p| \leq g$. *Floer-theoretically*, the lens space summand forces $HF^+(S^3_p(K), i)$ to be invariant, up to grading shift, under $i \mapsto i + |p|/a$ in the labelling of Spin^c structures by $\mathbb{Z}/|p|$. Using the Ozsváth–Szabó integer surgery formula we compute, for $2 \leq |p| \leq 2g-2$, a translation-invariant multiset τ of each such group: its size depends only on how many integers $s \equiv i \pmod{|p|}$ lie in $[-(g-1), g-1]$, and its elements depend only on these integers. The size count forces $|p| \mid 2g-1$, and a symmetry argument comparing the class 0 (whose τ has repeated entries) with any other class in the same orbit (whose τ has distinct entries) rules out $|p| \mid 2g-1$, $|p| < 2g-1$. Since $|p| \leq g$, this is a contradiction.

The cover bound (Proposition 6.1), the computation of τ (Proposition 7.3) and the arguments of Sections 4–8 are argued here; the inputs we use from the literature are collected, with references, in Section 3.

3 Conventions and cited results

We write $\mathbb{F} = \mathbb{Z}/2$. For a knot $K \subset S^3$, M_K is its exterior, μ and λ are its meridian and Seifert longitude on $T = \partial M_K$, and $g = g(K)$ is its genus. Prime decompositions are those of Kneser and Milnor [9, 11]: every closed oriented 3-manifold $Y \neq S^3$ is a connected sum $P_1\# \cdots \# P_k$ of prime manifolds $P_i \neq S^3$, and k and the P_i are unique (up to order and homeomorphism); k is the number of prime summands, and S^3 has none. A closed manifold with at least two prime summands is reducible, since a splitting sphere does not bound a ball.

Mirroring. If mK denotes the mirror of K, then $S^3_r(mK) \cong -S^3_{-r}(K)$, and reversing orientation does not change the number of prime summands. An L-space surgery is never at slope 0 (it has $b_1 = 1$). If K has an L-space surgery at a negative slope, then mK has one at a positive slope. Hence we may assume that K is a *positive L-space knot*, meaning that it has an L-space surgery at some positive slope. The surgery slope p studied below is then still an arbitrary integer of either sign, and both signs are treated.

We use the following known results.

Cited Theorem A (Gabai [1])**.** *If $K \subset S^3$ is a nontrivial knot, then $S^3_0(K)$ is irreducible.*

Cited Theorem B (Gordon–Luecke [4])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then r is an integer.*

Cited Theorem C (Howie [8]; the Scott–Wiegold conjecture)**.** *A free product of three nontrivial cyclic groups is not the normal closure of a single element.*

Cited Theorem D (Gordon–Luecke [5])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then one of its prime summands is a lens space (other than S^3 and $S^1 \times S^2$).*

We use only the weak consequence that some prime summand is an L-space L with $|H_1(L)| \geq 2$ (lens spaces are L-spaces [18]).

Cited Theorem E (Ni [15]; Ghiggini [2] for $g = 1$; Ozsváth–Szabó [18])**.** *An L-space knot K is fibred, and its fibre is a minimal genus Seifert surface F, of genus g, with one boundary component.*

(Ozsváth and Szabó [18] showed that the knot Floer homology of an L-space knot in its top Alexander grading is $\mathbb{F}$; Ni and Ghiggini showed that this characterises fibred knots.)

Cited Theorem F (Grushko [6], Neumann [14]; see also [10])**.** *If G_1, G_2 are finitely generated groups, then* $\operatorname{rank}(G_1 * G_2) = \operatorname{rank} G_1 + \operatorname{rank} G_2$*, where* rank *is the minimal number of generators.*

Together with van Kampen's theorem, $\pi_1(M_1 \# M_2) \cong \pi_1(M_1) * \pi_1(M_2)$, this gives $\operatorname{rank} \pi_1(M_1 \# M_2) = \operatorname{rank} \pi_1(M_1) + \operatorname{rank} \pi_1(M_2)$.

Cited Theorem G (Perelman [20, 21, 22]; see [12])**.** *A closed simply connected 3-manifold is homeomorphic to S^3.*

This can be avoided; see Remark 6.5.

Heegaard Floer input. An *L-space* is a rational homology sphere Y with $HF^+(Y, \mathfrak{t}) \cong \mathcal{T}^+ := \mathbb{F}[U, U^{-1}]/U \cdot \mathbb{F}[U]$ for every $\mathfrak{t} \in \operatorname{Spin}^c(Y)$ [18]. Let K be a positive L-space knot of genus $g \geq 1$ and $p \neq 0$ an integer.

(H1) *The integers V_s* ([19, 18]; the notation is that of Ni–Wu [16]). There are integers V_s, $s \in \mathbb{Z}$, with
$$V_s \geq 0, \qquad V_s - V_{s+1} \in \{0, 1\}, \qquad V_s = 0 \iff s \geq g, \qquad V_{-s} = V_s + s.$$
Put $H_s := V_{-s}$; thus $H_s = V_s + s$, and $H_s = 0 \iff s \leq -g$.

(H2) *The integer surgery mapping cone, plus version* (Ozsváth–Szabó [19]). The set $\operatorname{Spin}^c(S^3_p(K))$ is labelled by $\mathbb{Z}/|p|$. For each label i, $HF^+(S^3_p(K), i)$ is isomorphic, as a relatively $\mathbb{Z}$-graded $\mathbb{F}[U]$-module, to the homology of the mapping cone of
$$D = \sum_s (v_s + h_s) \colon \bigoplus_{s \equiv i} A^+_s \longrightarrow \bigoplus_{s \equiv i} B^+_s$$

(direct sums over $s \equiv i \pmod{|p|}$), where $v_s \colon A_s^+ \to B_s^+$ and $h_s \colon A_s^+ \to B_{s+p}^+$ are homogeneous chain maps. For a positive L-space knot, $H_*(A_s^+) \cong H_*(B_s^+) \cong \mathcal{T}^+$, and on homology v_s is U^{V_s} and h_s is U^{H_s} [18, 19].

Why homology-level maps suffice. $\mathbb{F}[U]$ is a graded principal ideal domain, hence hereditary. So every complex of graded $\mathbb{F}[U]$-modules is quasi-isomorphic to its homology (it is formal), and in the derived category $\mathrm{Hom}(X, Z) = \mathrm{Hom}(H(X), H(Z)) \oplus \mathrm{Ext}^1(H(X), H(Z))$, with the appropriate degree shift.

> **Reviewer 2**
>
> Complexes are quasi-isomorphic to their homology over a PID is false ($\mathbb{Z}$ is a PID).

The target homology $\bigoplus \mathcal{T}^+$ is injective: $\mathcal{T}^+$ is divisible, $\mathbb{F}[U]$ is Noetherian, and over a Noetherian ring direct sums of injective modules are injective. So the Ext^1 term vanishes, and the morphism D is determined up to isomorphism in the derived category by its homology map. The mapping cone of a morphism is determined up to isomorphism by the morphism. Hence the homology of the cone, with its relative grading, can be computed from the homology-level maps U^{V_s} and U^{H_s}.

(H3) *Affine labelling* [19]. The labelling $\mathrm{Spin}^c(S_p^3(K)) \leftrightarrow \mathbb{Z}/|p|$ is affine over $H_1(S_p^3(K)) = \mathbb{Z}/|p| = \langle [\mu] \rangle$, and $[\mu]$ acts by $i \mapsto i + 1$. (The sign convention does not matter below.)

(H4) *Connected sums* (Ozsváth–Szabó [17]). $\mathrm{Spin}^c(Y_1 \# Y_2) = \mathrm{Spin}^c(Y_1) \times \mathrm{Spin}^c(Y_2)$, equivariantly for $H_1(Y_1) \oplus H_1(Y_2)$, and $CF^-(Y_1 \# Y_2, (\mathfrak{t}_1, \mathfrak{t}_2)) \simeq CF^-(Y_1, \mathfrak{t}_1) \otimes_{\mathbb{F}[U]} CF^-(Y_2, \mathfrak{t}_2)$. If L is an L-space then $CF^-(L, \mathfrak{t}) \simeq \mathbb{F}[U]$, so $HF^+(L \# Y'', (\mathfrak{t}, \mathfrak{t}'')) \cong HF^+(Y'', \mathfrak{t}'')$ as relatively graded $\mathbb{F}[U]$-modules, up to a shift. Lens spaces are L-spaces [18].

4 Reduction

Suppose $S_r^3(K)$ has at least three prime summands; we derive a contradiction.

If K is the unknot, its surgeries are S^3, $S^1 \times S^2$ and lens spaces, each with at most one prime summand. So K is nontrivial. Slope ∞ gives S^3, and slope 0 gives an irreducible manifold by Theorem A. By Theorem B, a non-integral slope also gives an irreducible manifold. An irreducible closed 3-manifold has at most one prime summand.

So $r = p$ is a nonzero integer. After mirroring (Section 3), K is a nontrivial positive L-space knot of genus $g \geq 1$, and $p \neq 0$ has either sign. Write

$$Y = S_p^3(K) = P_1 \# \cdots \# P_k, \qquad P_i \text{ prime}, \ k \geq 3.$$

$H_1(Y) = \mathbb{Z}/|p|$ is finite. So no P_i is $S^1 \times S^2$, and each $H_1(P_i)$ is a direct summand of $\mathbb{Z}/|p|$, hence finite cyclic.

5 Two topological consequences

Lemma 5.1 (a homology sphere summand)**.** *Some P_i is an integer homology sphere $B \neq S^3$.*

> **Reviewer 2**
>
> A better statement for what this actually proves would be something like "All but at most 2 of the P_i are nontrivial integer homology spheres."

Proof. $\pi_1(M_K)$ is normally generated by μ, and $\pi_1(Y)$ is a quotient of it, so $\pi_1(Y)$ is normally generated by one element. Suppose three summands, say P_1, P_2, P_3, had nontrivial H_1. Then $\pi_1(Y) = \pi_1(P_1) * \cdots * \pi_1(P_k)$ maps onto $\pi_1(P_1) * \pi_1(P_2) * \pi_1(P_3)$, by killing the other factors, and then onto $H_1(P_1) * H_1(P_2) * H_1(P_3)$. This target is a free product of three nontrivial cyclic groups, and it is normally generated by the image of one element. This contradicts Theorem C. So at most two summands have nontrivial H_1. Since $k \geq 3$, some P_i has $H_1(P_i) = 0$, and $P_i \neq S^3$ because it is a prime summand. □

Write $Y \cong A \# B$, where B is this homology sphere and A is the connected sum of the other summands.

Lemma 5.2 (an L-space summand)**.** *$Y \cong L \# Y''$, where L is a lens space with $a := |H_1(L)| \geq 2$. Hence a divides $|p|$ and $|p| \geq 2$.*

Proof. Y is reducible since $k \geq 2$. By Theorem D, some prime summand is a lens space $L \neq S^3, S^1 \times S^2$, so $a \geq 2$. Also $H_1(L)$ is a direct summand of $\mathbb{Z}/|p|$. □

6 The cover bound

Proposition 6.1 (cover bound)**.** *Let $K \subset S^3$ be fibred, with fibre F of genus g and one boundary component, and let $p \neq 0$ be an integer. If $S^3_p(K) \cong A \# B$ with B an integer homology sphere other than S^3, then, with $N := |p|$ and $\tilde{A}$ the connected N-fold cyclic cover of A defined below,*

$$|p| \cdot \operatorname{rank} \pi_1(B) + \operatorname{rank} \pi_1(\tilde{A}) \leq 2g, \qquad \text{and hence} \qquad |p| \leq g.$$

Setup. $Y = S^3_p(K) = M_K \cup_T V$, where the meridian disc of the solid torus V is bounded by $\alpha = p\mu + \lambda$ on $T = \partial M_K$. $H_1(M_K) = \mathbb{Z}\langle\mu\rangle$; the class of a loop is its algebraic intersection number with F, and $\lambda = 0$ in $H_1(M_K)$. So $H_1(Y) = \mathbb{Z}/N$. Let $\varphi\colon \pi_1(Y) \to H_1(Y) = \mathbb{Z}/N$ be the abelianisation, which is onto, and let $q\colon Y_N \to Y$ be the connected N-fold cyclic cover with group $\ker \varphi$.

Lemma 6.2 (lower bound)**.** *$Y_N \cong \tilde{A} \# B \# \cdots \# B$ with N copies of B, and* $\operatorname{rank} \pi_1(Y_N) = \operatorname{rank} \pi_1(\tilde{A}) + N \operatorname{rank} \pi_1(B)$ *with* $\operatorname{rank} \pi_1(B) \geq 2$.

Proof. 1. Let S be the splitting sphere, so $Y = A^\circ \cup_S B^\circ$, where $^\circ$ means that one open ball is removed.

2. $\pi_1(Y) = \pi_1(A) * \pi_1(B)$, and $H_1(A) = H_1(Y) \cong \mathbb{Z}/N$ because $H_1(B) = 0$.

3. The restriction of φ to $\pi_1(B^\circ) = \pi_1(B)$ factors through $H_1(B) = 0$. So $q^{-1}(B^\circ)$ is N disjoint copies of B°, each mapped homeomorphically onto B°.

4. S is simply connected, so $q^{-1}(S)$ is N spheres.

5. The restriction of φ to $\pi_1(A^\circ) = \pi_1(A)$ is the surjection $\pi_1(A) \to H_1(A) \cong \mathbb{Z}/N$. So $q^{-1}(A^\circ)$ is connected: it is the induced N-fold cyclic cover $\tilde{A}$ of A with N balls removed (the removed ball lifts to N balls).

6. Therefore Y_N is $\tilde{A}$ minus N balls, with a copy of B° glued to each boundary sphere. The gluing graph is a star, hence a tree, so $Y_N \cong \tilde{A} \# B \# \cdots \# B$ with N copies of B. (If $N = 1$ the cover is trivial and $\tilde{A} = A$.)

7. By van Kampen and Theorem F, $\operatorname{rank}\pi_1(Y_N) = \operatorname{rank}\pi_1(\tilde{A}) + N\operatorname{rank}\pi_1(B)$.

8. $\pi_1(B)$ is perfect, since $H_1(B) = 0$, and nontrivial, by Theorem G, since $B \neq S^3$. A nontrivial cyclic group has nontrivial abelianisation, so $\pi_1(B)$ is not cyclic, and $\operatorname{rank}\pi_1(B) \geq 2$. □

Lemma 6.3 (upper bound)**.** $\operatorname{rank}\pi_1(Y_N) \leq 2g$.

Proof. 1. *The exterior lifts to a mapping torus.* On $\pi_1(M_K)$, φ is the composite $\pi_1(M_K) \to H_1(M_K) = \mathbb{Z} \to \mathbb{Z}/N$ with $\mu \mapsto 1$. So $E := q^{-1}(M_K)$ is connected and is the N-fold cyclic cover dual to F. It is the mapping torus of h^N, where h is the monodromy, and it fibres over S^1 with fibre $\tilde{F}$, a homeomorphic lift of F. So $\pi_1(E) = \pi_1(\tilde{F}) \rtimes \langle t \rangle$, and the fibration map $\pi\colon \pi_1(E) \to \mathbb{Z}$ has kernel $\pi_1(\tilde{F})$ and sends $t \mapsto 1$.

2. *The boundary lifts to one torus.* $\mu \in \pi_1(T)$ maps onto a generator of $\mathbb{Z}/N$. So $\tilde{T} := q^{-1}(T)$ is one torus, and $q|\colon \tilde{T} \to T$ has degree N.

3. *The solid torus lifts to one solid torus.* μ meets α once, so μ is isotopic in V to the core of V. Hence $\pi_1(V)$ maps onto $\mathbb{Z}/N$, and $\tilde{V} := q^{-1}(V)$ is one solid torus with $\partial\tilde{V} = \tilde{T}$.

4. *The filling curve.* The meridian disc of V lifts to N discs. Let $\beta \subset \tilde{T}$ be the boundary of one of them. Then q maps β homeomorphically onto α, so $q_*[\beta] = [\alpha]$, and $Y_N = E \cup \tilde{V}$ is the Dehn filling of E along β.

5. *Intersection numbers.* For a degree-N covering $f\colon \tilde{T} \to T$ of tori, $f_*x \cdot f_*y = N(x \cdot y)$ for $x, y \in H_1(\tilde{T})$. This follows from the transfer $f^!$: $f_*x \cdot z = x \cdot f^!z$ and $f^!f_* = N$, since deck transformations of a torus act trivially on H_1. Now $\tilde{F} \cap \tilde{T} = \partial\tilde{F}$ is a single curve $\tilde{\lambda}$ with $q_*[\tilde{\lambda}] = [\lambda]$. Hence
$$N(\beta \cdot \tilde{\lambda}) = \alpha \cdot \lambda = (p\mu + \lambda) \cdot \lambda = \pm p,$$
so $\beta \cdot \tilde{\lambda} = \pm 1$. This holds for either sign of p.

6. *β goes once around.* On $\pi_1(\tilde{T})$, the map π is algebraic intersection with the fibre boundary $\tilde{\lambda}$. So $\pi(\beta) = \pm 1$, and β represents $t^{\pm 1}w$ for some $w \in \pi_1(\tilde{F})$, for any choice of basepoint path.

7. *Conclusion.* By van Kampen, $\pi_1(Y_N) = \pi_1(E)/\langle\langle\beta\rangle\rangle$. In this quotient $t = w^{\mp 1}$ lies in the image of $\pi_1(\tilde{F})$. So $\pi_1(Y_N)$ is generated by the image of $\pi_1(\tilde{F})$, which is free of rank $2g$, and $\operatorname{rank}\pi_1(Y_N) \leq 2g$. □

Proof of Proposition 6.1. Combining Lemmas 6.2 and 6.3: $2N \leq N\operatorname{rank}\pi_1(B) \leq N\operatorname{rank}\pi_1(B) + \operatorname{rank}\pi_1(\tilde{A}) = \operatorname{rank}\pi_1(Y_N) \leq 2g$, so $|p| = N \leq g$. □

Remark 6.4. The bound is attained: for $T(2,3)$ and $p = 1$, Y is the Poincaré sphere and $2 = 2$. For a nontrivial fibred knot J and $p \geq 2$, the cable $C_{p,1}(J)$ is fibred of genus $p\,g(J) \geq p$, and its p-surgery is $L(p,1)\#S^3_{1/p}(J)$ [3], in agreement with the bound. (This cable is not an L-space knot; Proposition 6.1 needs only fibredness.)

Remark 6.5 (avoiding Theorem G)**.** Replace rank by Heegaard genus h. Since $\beta \cdot \tilde{\lambda} = \pm 1$, Y_N is the closed manifold of an open book with page $\tilde{F}$, so $h(Y_N) \leq 2g$. By Haken's lemma [7], $h(Y_N) = h(\tilde{A}) + Nh(B)$. Finally $h(B) \geq 2$, since the closed 3-manifolds of Heegaard genus at most 1 are S^3, $S^1 \times S^2$ and lens spaces. This again gives $N \leq g$.

7 Heegaard Floer input

In this section K is a nontrivial positive L-space knot of genus g, and $p \neq 0$ is an integer with $N := |p|$.

7.1 Coset invariance

Suppose $S^3_p(K) \cong L \# Y''$, where L is an L-space with $a = |H_1(L)| \geq 2$ and Y'' is a closed oriented 3-manifold. If the homeomorphism reverses orientation, replace (L, Y'') by $(-L, -Y'')$; $-L$ is again an L-space of the same order. Put $b := N/a$. Then b is an integer and $b \not\equiv 0 \pmod N$.

Lemma 7.1 (coset invariance)**.** *For every label i, $HF^+(S^3_p(K), i) \cong HF^+(S^3_p(K), i+b)$ as relatively graded $\mathbb{F}[U]$-modules, up to a degree shift.*

Proof. $H_1(L)$ is a subgroup of order a of $\mathbb{Z}/N = \langle [\mu] \rangle$, so it is $\langle b[\mu] \rangle$. By (H4), the Spin^c structures with a fixed Y''-component $\mathfrak{t}''$ form one $H_1(L)$-orbit, and all of them have $HF^+ \cong HF^+(Y'', \mathfrak{t}'')$ up to shift. By (H3), the $H_1(L)$-orbits are exactly the cosets $i + b\mathbb{Z}$ of labels. □

7.2 The invariant τ

Definition 7.2. Let M be a graded $\mathbb{F}[U]$-module that is finite-dimensional in each degree. Record degrees by *level*, meaning half the grading, so that U lowers the level by 1. Define $\tau(M)$ to be the multiset of levels k, where k appears with multiplicity $\dim(M_k / U \cdot M_{k+1})$.

τ is an isomorphism invariant, and shifting the grading translates τ. If $M = \mathcal{T}^+ \oplus \bigoplus_l \mathbb{F}[U]/U^{e_l}$, then $\tau(M)$ is the multiset of the top levels of the finite summands, because U maps onto $\mathcal{T}^+$; in particular $|\tau(M)|$ is the number of finite summands. Clearly $\tau(M \oplus M') = \tau(M) \cup \tau(M')$.

7.3 The level model and the structure of each class

Fix a class $c \in \mathbb{Z}/N$. Let $s_1 < s_2 < \cdots < s_m$ be the integers $s \equiv c \pmod N$ with $|s| \leq g - 1$. Consecutive ones differ by N, and

$$m = n_c := \#\{s \equiv c \pmod N : \ |s| \leq g - 1\}.$$

Level model. By (H2) we may compute with the homology-level maps U^{V_s} and U^{H_s} between copies of $\mathcal{T}^+$.

- Record each tower $\mathcal{T}^+$ by its *bottom level*.
- Regrade each B-tower so that the cone maps preserve levels. This is possible: in one class the A's and B's form a connected graph (each B is joined to its at most two adjacent A's), and the relative grading of the cone is homogeneous. A map U^e from a tower with bottom x then lands in a tower with bottom $x + e$.
- Under this regrading, an A-element at level k maps to the B-element at level k in each adjacent B that is *present at level k*, meaning that its bottom is $\leq k$. All exponents are ≥ 0, so every present B has its adjacent A's present.
- So at level k the cone map is the coboundary $\delta_k \colon \mathbb{F}^{\text{vertices}} \to \mathbb{F}^{\text{edges}}$ of a graph G_k. Its vertices are the A's with bottom $\leq k$, and its edges are the B's with bottom $\leq k$, each B joining its adjacent A's.

- U acts on cochains by restriction from G_{k+1} to G_k.
- The cone has two terms and D commutes with U, so $H(\text{cone}) = \ker D \oplus \operatorname{coker} D$ as graded $\mathbb{F}[U]$-modules.

The true relative grading is 2·level+constant on each side, because both satisfy the same constraints on a connected graph. So τ, computed in levels, is the invariant τ above up to translation.

Proposition 7.3 (structure of a class)**.** *Let c and $s_1 < \cdots < s_m$ be as above.*

(A) Let $0 < p \le 2g-2$ and $m \ge 1$. Then $\tau(HF^+(S^3_p(K), c)) = \{y - 1 : \ y \in Y_c\}$, where Y_c is the multiset of the $m-1$ interior peak *levels*

$$y_j = \frac{\mu_j^2 - p^2/4}{2p}, \qquad \mu_j = s_j + \frac{p}{2} \qquad (1 \le j \le m-1).$$

In particular $|\tau| = m - 1$.

(B) Let $p = -N < 0$ and $m \ge 1$. Let Π_c be the multiset of the $m+1$ peak *levels*

$$b(t) = -\frac{u^2 - N^2/4}{2N}, \qquad u = t + \frac{N}{2}, \qquad t \in \{s_1 - N,\, s_1,\, s_2, \ldots, s_m\},$$

and let M be its maximum. Then $\tau(HF^+(S^3_p(K), c))$ is $\{y - 1 : \ y \in \Pi_c\}$ with one copy of $M - 1$ removed. In particular $|\tau| = m$.

Neither multiset depends on K.

Proof of (A). In class c the A's and B's form one infinite path $\cdots A_{s-p} - B_s - A_s - B_{s+p} - \cdots$. Writing $x(s)$ for the bottom of A_s, the bottoms must satisfy $x(s+p) - x(s) = H_s - V_{s+p}$, since B_{s+p} receives U^{H_s} from A_s and $U^{V_{s+p}}$ from A_{s+p}. One solution is

$$x(s) = \frac{s(s-p)}{2p} - V_s;$$

indeed $x(s+p) - x(s) = s - V_{s+p} + V_s = H_s - V_{s+p}$, using $H_s = V_s + s$. The B between s and $s+p$ has bottom

$$y = x(s) + H_s = \frac{s(s+p)}{2p} = \frac{\mu^2 - p^2/4}{2p}, \qquad \mu = s + \frac{p}{2}.$$

We have $x(s) \to +\infty$ as $|s| \to \infty$: for $s \ge g$, $V_s = 0$, and for $s < 0$, $V_s = V_{-s} + |s| \le g + |s|$. So the quadratic term dominates, and each G_k is a finite subforest of a path.

The coboundary of a finite forest is onto. So D is surjective and $HF^+ \cong \ker D$. At level k, $\ker \delta_k$ is the space of functions on the components of G_k, and U is restriction of functions. Hence $\dim(M_k/U \cdot M_{k+1})$ is the number of *merges*: the number of components of G_k minus the number of components of G_{k+1} that contain a vertex of G_k.

Edges with a zero exponent. $V_s = 0$ iff $s \ge g$, and $H_s = 0$ iff $s \le -g$. A B with both exponents 0 would be B_{s+p} with $s \le -g$ and $s + p \ge g$, so $p \ge 2g$, which is excluded. A B with exactly one exponent 0 appears at the same level as its endpoint A_e on that side. That endpoint is new at that level. Its other edge has a positive exponent at A_e, since $V_e = H_e = 0$ would need $e \ge g$ and $e \le -g$; so that other edge appears strictly later. The far endpoint of B is old, since B's exponent there is positive. So at this level B attaches the new vertex A_e to one old component, and merges no old components.

Edges with both exponents ≥ 1. The edge B_{s+p} between A_s and A_{s+p} has exponents $H_s \geq 1$ (iff $s \geq -(g-1)$) and $V_{s+p} \geq 1$ (iff $s+p \leq g-1$). So these are exactly the $m-1$ edges between consecutive s_j, s_{j+1}: the interior peaks. Each appears strictly after both of its endpoints, and since G is a path, each one joins two distinct components of the old graph.

So the merges from level k to level $k+1$ are the interior peaks at level $k+1$. Hence $\tau = \{y-1 : y \text{ an interior peak}\}$ and $|\tau| = m-1$. □

Proof of (B). Here $h_s \colon A_s \to B_{s-N}$. Let C be the class-c cone.

Truncation, step 1. Let $P = \{A_s : s < g\} \cup \{B_t : t < g\}$. It is a subcomplex: for $s < g$, A_s maps to B_s and B_{s-N}, and both indices are $< g$. The quotient $Q = C/P = \{A_s, B_s : s \geq g\}$ has differential $D_Q = v_s + h_s$, where $v_s \colon A_s \to B_s$ is the identity since $V_s = 0$, and h_s lowers the index by N and is present only while $s - N \geq g$. On finitely supported chains, the identity plus a part that lowers the index, with indices bounded below, is bijective: invert it by a finite Neumann series. So Q is acyclic and $H(C) \cong H(P)$.

Truncation, step 2. Let $P' = \{A_s : |s| < g\} \cup \{B_t : -g-N < t < g\}$. It is a subcomplex of P: for $|s| < g$ we have $s - N > -g - N$. The quotient $R_0 = P/P' = \{A_s : s \leq -g\} \cup \{B_t : t \leq -g-N\}$ has differential $h_s + v_s$, where $h_s \colon A_s \to B_{s-N}$ is the identity since $H_s = 0$, and $v_s \colon A_s \to B_s$ is present only while $s \leq -g - N$. Relative to the pairing $A_s \leftrightarrow B_{s-N}$, the second part raises the B-index by N, and indices are bounded above. So it is again identity plus nilpotent on finitely supported chains, and R_0 is acyclic. Hence $H(C) \cong H(P')$.

Shape of P'. The class-c integers in $(-g-N, g)$ are $s_1 - N, s_1, \ldots, s_m$, because $s_1 - N \leq -g < s_1$ and $s_m < g \leq s_m + N$. So P' is the zigzag

$$B_{s_1-N} - A_{s_1} - B_{s_1} - A_{s_2} - \cdots - A_{s_m} - B_{s_m}.$$

Every exponent on it is ≥ 1: $V_{s_j} \geq 1$ since $s_j \leq g-1$, and $H_{s_j} = V_{-s_j} \geq 1$ since $-s_j \leq g-1$. The two end B's each have one A-neighbour. Treat them as edges to a *ground* vertex, at which cochains take the value 0. Then $G \cup \{\text{ground}\}$ is a single cycle with $m+1$ edges.

Levels. Put $x(s) = -\frac{s(s+N)}{2N} - V_s$ and $b(t) = -\frac{t(t+N)}{2N} = -\frac{u^2 - N^2/4}{2N}$, $u = t + \frac{N}{2}$. Then B_s has bottom $x(s) + V_s = b(s)$, and since $H_s = V_s + s$,

$$x(s) + H_s = b(s) + s = \frac{-s^2 - sN + 2sN}{2N} = -\frac{s(s-N)}{2N} = b(s-N),$$

as required for B_{s-N}.

The cokernel part. $\operatorname{coker} \delta_k = H^1(G_k \cup \text{ground}, \text{ground})$, which is $H^1(G_k \cup \text{ground})$ because the ground is one point. This is 0 until the cycle closes, at the maximal peak level M. From then on it is $\mathbb{F}$, and U (restriction of cochains) is an isomorphism. So the cokernel part is a single tower and contributes nothing to τ.

The kernel part. $\ker \delta_k$ is the space of functions on the components of G_k that do not contain the ground. A vertex that is new at level $k+1$ has its edges strictly higher, since all exponents are ≥ 1. So it is isolated at level $k+1$, and the new edges at level $k+1$ join only old vertices or the ground. Hence $\dim(\ker_k / U \cdot \ker_{k+1})$ is the number of merges among old components, including merges into the ground component. That number is $\#(\text{new edges at level } k+1) - \Delta b_1$, and b_1 increases (by 1) only at level M. So $\tau = \{y - 1 : y \in \Pi_c\}$ with one copy of $M-1$ removed, and $|\tau| = (m+1) - 1 = m$. □

7.4 Divisibility

Proposition 7.4. *Suppose $2 \leq |p| \leq 2g-2$ and $S^3_p(K) \cong L \# Y''$ with L an L-space of order $a \geq 2$. Then $|p|$ divides $2g-1$.*

Proof. Write $2g-1 = QN + r$ with $0 \le r < N$. Then $Q \ge 1$, since $N \le 2g-2$. The integers in $[-(g-1), g-1]$ are $2g-1$ consecutive integers, so $n_i = Q+1$ on a set S of r cyclically consecutive residues, and $n_i = Q$ elsewhere. In particular every $n_i \ge 1$.

By Proposition 7.3, $|\tau(HF^+(S^3_p(K), i))|$ equals $n_i - 1$ if $p > 0$ and n_i if $p < 0$; either way it is a strictly increasing function of n_i. By Lemma 7.1 and the invariance of τ up to translation, $|\tau|$ is invariant under $i \mapsto i + b$ with $b \not\equiv 0$. So S is invariant under translation by b.

If $0 < r < N$, then S is a proper nonempty cyclic interval in $\mathbb{Z}/N$. It has exactly one start point (an $x \in S$ with $x - 1 \notin S$), and a translation preserving S must fix that start point. That is impossible for $b \not\equiv 0$. So $r = 0$, i.e. N divides $2g-1$. □

7.5 The key contradiction

Theorem 7.5. *Let K be a nontrivial positive L-space knot of genus g, and p a nonzero integer with $|p| \ge 2$, $|p|$ dividing $2g-1$ and $|p| < 2g-1$. Then $S^3_p(K)$ is not homeomorphic to $L \# Y''$ for any L-space L with $|H_1(L)| \ge 2$ and any closed oriented 3-manifold Y''.*

Proof. Let $N := |p|$ and $n := (2g-1)/N$. Both are odd, $N \ge 3$ and $n \ge 3$. In particular $|p| \le (2g-1)/3 \le 2g-2$, so Proposition 7.3 applies. Suppose such a splitting exists, and let b be as in Lemma 7.1.

(i) The comparison. Let t be the representative of b with $0 < |t| \le (N-1)/2$; it exists because N is odd and $b \not\equiv 0$. By Lemma 7.1 and the invariance of τ:

$$(\star) \qquad \tau \text{ of class } 0 \text{ and } \tau \text{ of class } t \text{ agree up to a translation.}$$

(ii) Arithmetic. For $|c| \le (N-1)/2$, the integers $s \equiv c$ with $|s| \le g-1$ are exactly $c + kN$ for $|k| \le (n-1)/2$. Indeed $|c + kN| \le (N-1)/2 + (n-1)N/2 = g-1$, while $|k| \ge (n+1)/2$ gives $|c + kN| \ge (nN+1)/2 > g-1$. So $m = n$ in every class.

Key fact. Let $u = c + (k + \frac{1}{2})N$ and $u' = c + (k' + \frac{1}{2})N$ with $k \ne k'$. If $u^2 = u'^2$, then $u = -u'$, so $2c + (k + k' + 1)N = 0$. Hence $N \mid 2c$, so $N \mid c$ because N is odd, and $c = 0$. So for $c = t$, distinct k give distinct values of u^2, hence distinct levels in both formulas of Proposition 7.3.

(iii) Case $p > 0$. Class 0: $s_j = (j - \frac{n+1}{2})N$, so the interior peaks have $\mu_j = s_j + N/2 = (j - \frac{n}{2})N$ for $1 \le j \le n-1$. These are $\pm N/2, \pm 3N/2, \ldots, \pm(n-2)N/2$: nonzero $\pm$ pairs. So every element of τ has even multiplicity, and τ is nonempty since $n - 1 \ge 2$. *Class t:* $\mu_j = t + (j - \frac{n+1}{2} + \frac{1}{2})N$, and by (ii) the $n-1$ levels are distinct. A translate of a nonempty multiset in which every multiplicity is even never has all multiplicities equal to 1. This contradicts $(\star)$.

(iv) Case $p < 0$. Class 0: the peaks are $t' = kN$ for $k = -\frac{n+1}{2}, \ldots, \frac{n-1}{2}$, so $u = t' + N/2$ runs over $\pm N/2, \pm 3N/2, \ldots, \pm nN/2$, each value of $|u|$ exactly twice. The maximal level M comes from $|u| = N/2$. Removing one copy of $M - 1$ leaves the levels for $|u| = 3N/2, \ldots, nN/2$ still doubled, and these exist since $n \ge 3$. So τ has a repeated element. *Class t:* by (ii) the $n+1$ peak levels are distinct, so the n elements of τ are distinct. This contradicts $(\star)$. □

8 Proof of the Main Theorem

Assume, as in Section 4, that K is a nontrivial positive L-space knot of genus g, $p \ne 0$, and $Y = S^3_p(K)$ has $k \ge 3$ prime summands.

1. By Lemma 5.1, $Y \cong A \# B$ with B an integer homology sphere, $B \ne S^3$.

2. By Lemma 5.2, $Y \cong L \# Y''$ with L a lens space, hence an L-space, of order $a \ge 2$. So $|p| \ge 2$.

3. K is fibred with fibre of genus g and one boundary component (Theorem E). Proposition 6.1 gives $|p| \leq g$. If $g = 1$, this contradicts $|p| \geq 2$. So $g \geq 2$, and $2 \leq |p| \leq g \leq 2g - 2$.

4. By Proposition 7.4, applied to the splitting of step 2, $|p|$ divides $2g - 1$. Also $|p| \leq g < 2g - 1$.

5. By Theorem 7.5, the splitting of step 2 cannot exist. This is a contradiction.

So $S^3_r(K)$ has at most two prime summands, for every L-space knot K and every slope r. □

9 Remarks

- For a fibred knot, a reducible integral surgery $S^3_p(K)$ with an integer homology sphere summand $B \neq S^3$ requires $|p| \leq g$ (Proposition 6.1). This uses only fibredness.
- Granting Theorem D, the argument of Section 8 shows more: a *reducible* surgery on an L-space knot never has a prime summand that is an integer homology sphere other than S^3, since steps 2–5 use only the homology sphere summand and the lens space summand. (Irreducible surgeries can be homology spheres, e.g. $S^3_{+1}(T(2,3))$ is the Poincaré sphere.) This agrees with the known examples, cables at the cabling slope, whose summands are $L(a,b)$ and $S^3_{b/a}(J)$.

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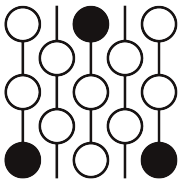
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Review of Problem 6 – Siege

First Proof Third Batch
Reviewer 3

October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- -0.5: major errors
- -1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: N/A

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- -1: very poor exposition

Score: -1.

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- -1: blatant plagiarism or hallucinations

Score: -1.

The solution, which is largely plagiarized, appears to be correct. I am basing this assessment on the fact that the plagiarized paper is correct. The main new ideas are in Section 6 and the arguments seem right, though they are very poorly written (even beyond the fact the proofs of the key lemmas are lists of bullet points).

The exposition is abominable. Here is Lemma 5.1 in its entirety: "Some P_i is an integer homology sphere $B \neq S^3$." It comes at the start of a section, so one is supposed to infer the meaning of P_i from the previous section. No hypotheses are stated.

The above is not the worst offense. There are long chunks of text that use notation which has not been set up. Even the parts that are possible to follow with reasonable effort are often too convoluted. As a comparatively minor point, proofs shouldn't be just a bunch of bullet points.

Section 7 is a case of blatant plagiarism. It takes ideas of Hom-Lidman-Zufelt but does not acknowledge this in any way. There is also a lemma stolen from Howie who is not cited. There is at least one instance where the wrong source is cited. There is also the smaller but significant problem that citations are misplaced within the text.

2 Submission with inline comments

Problem

Does a 3-manifold obtained by Dehn surgery on an L-space knot have at most two prime summands?

Result

Throughout, an *L-space knot* is a knot $K \subset S^3$ such that $S^3_r(K)$ is an L-space for some slope $r \in \mathbb{Q}$. The answer is **yes**, for every slope.

> **Reviewer 3**
> The above sentence doesn't need to be there. Overall, the exposition up to here is in very poor form.

Main Theorem. *Let $K \subset S^3$ be an L-space knot and $r \in \mathbb{Q} \cup \{\infty\}$. Then $S^3_r(K)$ has at most two prime summands.*

The bound is sharp. By Moser [13], $S^3_6(T(2,3)) \cong L(2,1)\#L(3,2)$, which has two prime summands. More generally, if a cable $C_{a,b}(J)$ (with winding number a) is an L-space knot, then its surgery at the cabling slope ab is $L(a,b)\#S^3_{b/a}(J)$ [3].

Overview of the proof

Suppose $Y = S^3_r(K)$ has at least three prime summands. Classical results (Gabai; Gordon–Luecke)

> **Reviewer 3**
> need a precise reference

force $r = p$ to be a nonzero integer and K to be nontrivial. Since $\pi_1(Y)$ is normally generated by one element, Howie's solution of the Scott–Wiegold conjecture

> **Reviewer 3**
> state precisely and cite

shows that at most two summands have nontrivial first homology; hence some summand B is an integer homology sphere other than S^3. A theorem of Gordon–Luecke supplies a lens space summand L of order $a \geq 2$.

> **Reviewer 3**
> which theorem?

Two independent arguments then collide. *Topologically*, K is fibred (Ni, Ghiggini),

> **Reviewer 3**
> missing citation again

and in the $|p|$-fold cyclic cover of Y the summand B lifts to $|p|$ copies, while the fibred structure shows that the cover has fundamental group generated by $2g$ elements. Grushko's theorem then gives $|p| \leq g$. *Floer-theoretically*, the lens space summand forces $HF^+(S^3_p(K), i)$ to be invariant, up to grading shift, under $i \mapsto i + |p|/a$ in the labelling of Spin^c structures by $\mathbb{Z}/|p|$. Using the Ozsváth–Szabó integer surgery formula we compute, for $2 \leq |p| \leq 2g-2$, a translation-invariant multiset τ of each such group: its size depends only on how many integers $s \equiv i \pmod{|p|}$ lie in

$[-(g-1), g-1]$, and its elements depend only on these integers. The size count forces $|p| \mid 2g-1$, and a symmetry argument comparing the class 0 (whose τ has repeated entries) with any other class in the same orbit (whose τ has distinct entries) rules out $|p| \mid 2g-1$, $|p| < 2g-1$. Since $|p| \le g$, this is a contradiction.

> **Reviewer 3**
>
> I find the above paragraph very hard to follow.

The cover bound (Proposition 6.1), the computation of τ (Proposition 7.3) and the arguments of Sections 4–8 are argued here; the inputs we use from the literature are collected, with references, in Section 3.

> **Reviewer 3**
>
> Ok, so maybe the correct citations appear later. Still, they should have been included in the above as well.

3 Conventions and cited results

We write $\mathbb{F} = \mathbb{Z}/2$. For a knot $K \subset S^3$, M_K is its exterior, μ and λ are its meridian and Seifert longitude on $T = \partial M_K$, and $g = g(K)$ is its genus. Prime decompositions are those of Kneser and Milnor [9, 11]: every closed oriented 3-manifold $Y \neq S^3$ is a connected sum $P_1 \# \cdots \# P_k$ of prime manifolds $P_i \neq S^3$, and k and the P_i are unique (up to order and homeomorphism); k is the number of prime summands, and S^3 has none. A closed manifold with at least two prime summands is reducible, since a splitting sphere does not bound a ball.

Mirroring. If mK denotes the mirror of K, then $S^3_r(mK) \cong -S^3_{-r}(K)$, and reversing orientation does not change the number of prime summands. An L-space surgery is never at slope 0 (it has $b_1 = 1$). If K has an L-space surgery at a negative slope, then mK has one at a positive slope. Hence we may assume that K is a *positive L-space knot*, meaning that it has an L-space surgery at some positive slope. The surgery slope p studied below is then still an arbitrary integer of either sign, and both signs are treated.

We use the following known results.

Cited Theorem A (Gabai [1])**.** *If $K \subset S^3$ is a nontrivial knot, then $S^3_0(K)$ is irreducible.*

Cited Theorem B (Gordon–Luecke [4])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then r is an integer.*

Cited Theorem C (Howie [8]; the Scott–Wiegold conjecture)**.** *A free product of three nontrivial cyclic groups is not the normal closure of a single element.*

Cited Theorem D (Gordon–Luecke [5])**.** *If $K \subset S^3$ is nontrivial and $S^3_r(K)$ is reducible, then one of its prime summands is a lens space (other than S^3 and $S^1 \times S^2$).*

> **Reviewer 3**
>
> The wrong Gordon–Luecke paper is cited for this result.

We use only the weak consequence that some prime summand is an L-space L with $|H_1(L)| \ge 2$ (lens spaces are L-spaces [18]).

Cited Theorem E (Ni [15]; Ghiggini [2] for $g = 1$; Ozsváth–Szabó [18])**.** *An L-space knot K is fibred, and its fibre is a minimal genus Seifert surface F, of genus g, with one boundary component.*

> **Reviewer 3**
>
> The above theorem is very badly phrased. Why would you point out, in a theorem statement no less, that the Seifert surface for a knot has connected boundary? Also, is g the genus of K? If so, we're missing a reference to Neuwirth–Stallings.

(Ozsváth and Szabó [18] showed that the knot Floer homology of an L-space knot in its top Alexander grading is $\mathbb{F}$; Ni and Ghiggini showed that this characterises fibred knots.)

Cited Theorem F (Grushko [6], Neumann [14]; see also [10])**.** *If G_1, G_2 are finitely generated groups, then* $\operatorname{rank}(G_1 * G_2) = \operatorname{rank} G_1 + \operatorname{rank} G_2$*, where* rank *is the minimal number of generators.*

Together with van Kampen's theorem, $\pi_1(M_1 \# M_2) \cong \pi_1(M_1) * \pi_1(M_2)$, this gives $\operatorname{rank} \pi_1(M_1 \# M_2) = \operatorname{rank} \pi_1(M_1) + \operatorname{rank} \pi_1(M_2)$.

Cited Theorem G (Perelman [20, 21, 22]; see [12])**.** *A closed simply connected 3-manifold is homeomorphic to S^3.*

This can be avoided; see Remark 6.5.

Heegaard Floer input. An *L-space* is a rational homology sphere Y with $HF^+(Y, \mathfrak{t}) \cong \mathcal{T}^+ := \mathbb{F}[U, U^{-1}]/U \cdot \mathbb{F}[U]$ for every $\mathfrak{t} \in \operatorname{Spin}^c(Y)$ [18]. Let K be a positive L-space knot of genus $g \geq 1$ and $p \neq 0$ an integer.

(H1) *The integers V_s* ([19, 18]; the notation is that of Ni–Wu [16]). There are integers V_s, $s \in \mathbb{Z}$, with

$$V_s \geq 0, \qquad V_s - V_{s+1} \in \{0, 1\}, \qquad V_s = 0 \iff s \geq g, \qquad V_{-s} = V_s + s.$$

Put $H_s := V_{-s}$; thus $H_s = V_s + s$, and $H_s = 0 \iff s \leq -g$.

(H2) *The integer surgery mapping cone, plus version* (Ozsváth–Szabó [19]). The set $\operatorname{Spin}^c(S^3_p(K))$ is labelled by $\mathbb{Z}/|p|$. For each label i, $HF^+(S^3_p(K), i)$ is isomorphic, as a relatively $\mathbb{Z}$-graded $\mathbb{F}[U]$-module, to the homology of the mapping cone of

$$D = \sum_s (v_s + h_s) \colon \bigoplus_{s \equiv i} A^+_s \longrightarrow \bigoplus_{s \equiv i} B^+_s$$

(direct sums over $s \equiv i \pmod{|p|}$), where $v_s \colon A^+_s \to B^+_s$ and $h_s \colon A^+_s \to B^+_{s+p}$ are homogeneous chain maps. For a positive L-space knot, $H_*(A^+_s) \cong H_*(B^+_s) \cong \mathcal{T}^+$, and on homology v_s is U^{V_s} and h_s is U^{H_s} [18, 19].

> **Reviewer 3**
>
> This is unreadable. The notation is not defined. I don't know what I'm looking at. (Presumably the idea is to look up the notation in Ni-Wu; this is unacceptable.) At this point, if this was a paper submitted to a journal, I would read no further.

Why homology-level maps suffice. $\mathbb{F}[U]$ is a graded principal ideal domain, hence hereditary. So every complex of graded $\mathbb{F}[U]$-modules is quasi-isomorphic to its homology (it is formal), and in the derived category $\operatorname{Hom}(X, Z) = \operatorname{Hom}(H(X), H(Z)) \oplus \operatorname{Ext}^1(H(X), H(Z))$, with the appropriate degree shift. The target homology $\bigoplus \mathcal{T}^+$ is injective: $\mathcal{T}^+$ is divisible, $\mathbb{F}[U]$ is Noetherian, and over a Noetherian ring direct sums of injective modules are injective. So the Ext^1 term vanishes, and the morphism D is determined up to isomorphism in the derived category by its homology map. The mapping cone of a morphism is determined up to isomorphism by the morphism. Hence the homology of the cone, with its relative grading, can be computed from the homology-level maps U^{V_s} and U^{H_s}.

(H3) *Affine labelling* [19]. The labelling $\mathrm{Spin}^c(S^3_p(K)) \leftrightarrow \mathbb{Z}/|p|$ is affine over $H_1(S^3_p(K)) = \mathbb{Z}/|p| = \langle[\mu]\rangle$, and $[\mu]$ acts by $i \mapsto i+1$. (The sign convention does not matter below.)

(H4) *Connected sums* (Ozsváth–Szabó [17]). $\mathrm{Spin}^c(Y_1 \# Y_2) = \mathrm{Spin}^c(Y_1) \times \mathrm{Spin}^c(Y_2)$, equivariantly for $H_1(Y_1) \oplus H_1(Y_2)$, and $CF^-(Y_1 \# Y_2, (\mathfrak{t}_1, \mathfrak{t}_2)) \simeq CF^-(Y_1, \mathfrak{t}_1) \otimes_{\mathbb{F}[U]} CF^-(Y_2, \mathfrak{t}_2)$. If L is an L-space then $CF^-(L, \mathfrak{t}) \simeq \mathbb{F}[U]$, so $HF^+(L \# Y'', (\mathfrak{t}, \mathfrak{t}'')) \cong HF^+(Y'', \mathfrak{t}'')$ as relatively graded $\mathbb{F}[U]$-modules, up to a shift. Lens spaces are L-spaces [18].

4 Reduction

Suppose $S^3_r(K)$ has at least three prime summands; we derive a contradiction.

If K is the unknot, its surgeries are S^3, $S^1 \times S^2$ and lens spaces, each with at most one prime summand. So K is nontrivial. Slope ∞ gives S^3, and slope 0 gives an irreducible manifold by Theorem A. By Theorem B, a non-integral slope also gives an irreducible manifold. An irreducible closed 3-manifold has at most one prime summand.

So $r = p$ is a nonzero integer. After mirroring (Section 3), K is a nontrivial positive L-space knot of genus $g \geq 1$, and $p \neq 0$ has either sign. Write

$$Y = S^3_p(K) = P_1 \# \cdots \# P_k, \qquad P_i \text{ prime}, \ k \geq 3.$$

$H_1(Y) = \mathbb{Z}/|p|$ is finite. So no P_i is $S^1 \times S^2$, and each $H_1(P_i)$ is a direct summand of $\mathbb{Z}/|p|$, hence finite cyclic.

5 Two topological consequences

Lemma 5.1 (a homology sphere summand)**.** *Some P_i is an integer homology sphere $B \neq S^3$.*

> **Reviewer 3**
>
> What is the notation here? What are the hypotheses?

Proof. $\pi_1(M_K)$ is normally generated by μ, and $\pi_1(Y)$ is a quotient of it, so $\pi_1(Y)$ is normally generated by one element. Suppose three summands, say P_1, P_2, P_3, had nontrivial H_1. Then $\pi_1(Y) = \pi_1(P_1) * \cdots * \pi_1(P_k)$ maps onto $\pi_1(P_1) * \pi_1(P_2) * \pi_1(P_3)$, by killing the other factors, and then onto $H_1(P_1) * H_1(P_2) * H_1(P_3)$. This target is a free product of three nontrivial cyclic groups, and it is normally generated by the image of one element. This contradicts Theorem C. So at most two summands have nontrivial H_1. Since $k \geq 3$, some P_i has $H_1(P_i) = 0$, and $P_i \neq S^3$ because it is a prime summand. □

> **Reviewer 3**
>
> This is in the Howie paper and needs to be attributed

Write $Y \cong A \# B$, where B is this homology sphere and A is the connected sum of the other summands.

Lemma 5.2 (an L-space summand)**.** *$Y \cong L \# Y''$, where L is a lens space with $a := |H_1(L)| \geq 2$. Hence a divides $|p|$ and $|p| \geq 2$.*

Proof. Y is reducible since $k \geq 2$. By Theorem D, some prime summand is a lens space $L \neq S^3, S^1 \times S^2$, so $a \geq 2$. Also $H_1(L)$ is a direct summand of $\mathbb{Z}/|p|$. □

6 The cover bound

Proposition 6.1 (cover bound)**.** *Let $K \subset S^3$ be fibred, with fibre F of genus g and one boundary component, and let $p \neq 0$ be an integer. If $S^3_p(K) \cong A \# B$ with B an integer homology sphere other than S^3, then, with $N := |p|$ and $\tilde{A}$ the connected N-fold cyclic cover of A defined below,*

$$|p| \cdot \operatorname{rank} \pi_1(B) + \operatorname{rank} \pi_1(\tilde{A}) \leq 2g, \qquad \textit{and hence} \qquad |p| \leq g.$$

Setup. $Y = S^3_p(K) = M_K \cup_T V$, where the meridian disc of the solid torus V is bounded by $\alpha = p\mu + \lambda$ on $T = \partial M_K$. $H_1(M_K) = \mathbb{Z}\langle\mu\rangle$; the class of a loop is its algebraic intersection number with F, and $\lambda = 0$ in $H_1(M_K)$. So $H_1(Y) = \mathbb{Z}/N$. Let $\varphi\colon \pi_1(Y) \to H_1(Y) = \mathbb{Z}/N$ be the abelianisation, which is onto, and let $q\colon Y_N \to Y$ be the connected N-fold cyclic cover with group $\ker \varphi$.

Lemma 6.2 (lower bound)**.** *$Y_N \cong \tilde{A} \# B \# \cdots \# B$ with N copies of B, and* $\operatorname{rank} \pi_1(Y_N) = \operatorname{rank} \pi_1(\tilde{A}) + N \operatorname{rank} \pi_1(B)$ *with* $\operatorname{rank} \pi_1(B) \geq 2$.

Proof. 1. Let S be the splitting sphere, so $Y = A^\circ \cup_S B^\circ$, where $^\circ$ means that one open ball is removed.

2. $\pi_1(Y) = \pi_1(A) * \pi_1(B)$, and $H_1(A) = H_1(Y) \cong \mathbb{Z}/N$ because $H_1(B) = 0$.

3. The restriction of φ to $\pi_1(B^\circ) = \pi_1(B)$ factors through $H_1(B) = 0$. So $q^{-1}(B^\circ)$ is N disjoint copies of B°, each mapped homeomorphically onto B°.

4. S is simply connected, so $q^{-1}(S)$ is N spheres.

5. The restriction of φ to $\pi_1(A^\circ) = \pi_1(A)$ is the surjection $\pi_1(A) \to H_1(A) \cong \mathbb{Z}/N$. So $q^{-1}(A^\circ)$ is connected: it is the induced N-fold cyclic cover $\tilde{A}$ of A with N balls removed (the removed ball lifts to N balls).

6. Therefore Y_N is $\tilde{A}$ minus N balls, with a copy of B° glued to each boundary sphere. The gluing graph is a star, hence a tree, so $Y_N \cong \tilde{A} \# B \# \cdots \# B$ with N copies of B. (If $N = 1$ the cover is trivial and $\tilde{A} = A$.)

7. By van Kampen and Theorem F, $\operatorname{rank} \pi_1(Y_N) = \operatorname{rank} \pi_1(\tilde{A}) + N \operatorname{rank} \pi_1(B)$.

8. $\pi_1(B)$ is perfect, since $H_1(B) = 0$, and nontrivial, by Theorem G, since $B \neq S^3$. A nontrivial cyclic group has nontrivial abelianisation, so $\pi_1(B)$ is not cyclic, and $\operatorname{rank} \pi_1(B) \geq 2$. □

Lemma 6.3 (upper bound)**.** $\operatorname{rank} \pi_1(Y_N) \leq 2g$.

Proof. 1. *The exterior lifts to a mapping torus.* On $\pi_1(M_K)$, φ is the composite $\pi_1(M_K) \to H_1(M_K) = \mathbb{Z} \to \mathbb{Z}/N$ with $\mu \mapsto 1$. So $E := q^{-1}(M_K)$ is connected and is the N-fold cyclic cover dual to F. It is the mapping torus of h^N, where h is the monodromy, and it fibres over S^1 with fibre $\tilde{F}$, a homeomorphic lift of F. So $\pi_1(E) = \pi_1(\tilde{F}) \rtimes \langle t \rangle$, and the fibration map $\pi\colon \pi_1(E) \to \mathbb{Z}$ has kernel $\pi_1(\tilde{F})$ and sends $t \mapsto 1$.

2. *The boundary lifts to one torus.* $\mu \in \pi_1(T)$ maps onto a generator of $\mathbb{Z}/N$. So $\tilde{T} := q^{-1}(T)$ is one torus, and $q|\colon \tilde{T} \to T$ has degree N.

3. *The solid torus lifts to one solid torus.* μ meets α once, so μ is isotopic in V to the core of V. Hence $\pi_1(V)$ maps onto $\mathbb{Z}/N$, and $\tilde{V} := q^{-1}(V)$ is one solid torus with $\partial \tilde{V} = \tilde{T}$.

4. *The filling curve.* The meridian disc of V lifts to N discs. Let $\beta \subset \tilde{T}$ be the boundary of one of them. Then q maps β homeomorphically onto α, so $q_*[\beta] = [\alpha]$, and $Y_N = E \cup \tilde{V}$ is the Dehn filling of E along β.

5. *Intersection numbers.* For a degree-N covering $f\colon \tilde{T} \to T$ of tori, $f_*x \cdot f_*y = N(x \cdot y)$ for $x, y \in H_1(\tilde{T})$. This follows from the transfer $f^!$: $f_*x \cdot z = x \cdot f^!z$ and $f^!f_* = N$, since deck transformations of a torus act trivially on H_1. Now $\tilde{F} \cap \tilde{T} = \partial\tilde{F}$ is a single curve $\tilde{\lambda}$ with $q_*[\tilde{\lambda}] = [\lambda]$. Hence
$$N(\beta \cdot \tilde{\lambda}) = \alpha \cdot \lambda = (p\mu + \lambda) \cdot \lambda = \pm p,$$
so $\beta \cdot \tilde{\lambda} = \pm 1$. This holds for either sign of p.

6. *β goes once around.* On $\pi_1(\tilde{T})$, the map π is algebraic intersection with the fibre boundary $\tilde{\lambda}$. So $\pi(\beta) = \pm 1$, and β represents $t^{\pm 1}w$ for some $w \in \pi_1(\tilde{F})$, for any choice of basepoint path.

7. *Conclusion.* By van Kampen, $\pi_1(Y_N) = \pi_1(E)/\langle\langle\beta\rangle\rangle$. In this quotient $t = w^{\mp 1}$ lies in the image of $\pi_1(\tilde{F})$. So $\pi_1(Y_N)$ is generated by the image of $\pi_1(\tilde{F})$, which is free of rank $2g$, and $\operatorname{rank}\pi_1(Y_N) \le 2g$. □

Proof of Proposition 6.1. Combining Lemmas 6.2 and 6.3: $2N \le N\operatorname{rank}\pi_1(B) \le N\operatorname{rank}\pi_1(B) + \operatorname{rank}\pi_1(\tilde{A}) = \operatorname{rank}\pi_1(Y_N) \le 2g$, so $|p| = N \le g$. □

Remark 6.4. The bound is attained: for $T(2,3)$ and $p = 1$, Y is the Poincaré sphere and $2 = 2$. For a nontrivial fibred knot J and $p \ge 2$, the cable $C_{p,1}(J)$ is fibred of genus $p\,g(J) \ge p$, and its p-surgery is $L(p,1)\#S^3_{1/p}(J)$ [3], in agreement with the bound. (This cable is not an L-space knot; Proposition 6.1 needs only fibredness.)

Remark 6.5 (avoiding Theorem G)**.** Replace rank by Heegaard genus h. Since $\beta \cdot \tilde{\lambda} = \pm 1$, Y_N is the closed manifold of an open book with page $\tilde{F}$, so $h(Y_N) \le 2g$. By Haken's lemma [7], $h(Y_N) = h(\tilde{A}) + Nh(B)$. Finally $h(B) \ge 2$, since the closed 3-manifolds of Heegaard genus at most 1 are S^3, $S^1 \times S^2$ and lens spaces. This again gives $N \le g$.

7 Heegaard Floer input

In this section K is a nontrivial positive L-space knot of genus g, and $p \ne 0$ is an integer with $N := |p|$.

7.1 Coset invariance

Suppose $S^3_p(K) \cong L\#Y''$, where L is an L-space with $a = |H_1(L)| \ge 2$ and Y'' is a closed oriented 3-manifold. If the homeomorphism reverses orientation, replace (L, Y'') by $(-L, -Y'')$; $-L$ is again an L-space of the same order. Put $b := N/a$. Then b is an integer and $b \not\equiv 0 \pmod N$.

Lemma 7.1 (coset invariance)**.** *For every label i, $HF^+(S^3_p(K), i) \cong HF^+(S^3_p(K), i+b)$ as relatively graded $\mathbb{F}[U]$-modules, up to a degree shift.*

Proof. $H_1(L)$ is a subgroup of order a of $\mathbb{Z}/N = \langle[\mu]\rangle$, so it is $\langle b[\mu]\rangle$. By (H4), the Spin^c structures with a fixed Y''-component $\mathfrak{t}''$ form one $H_1(L)$-orbit, and all of them have $HF^+ \cong HF^+(Y'', \mathfrak{t}'')$ up to shift. By (H3), the $H_1(L)$-orbits are exactly the cosets $i + b\mathbb{Z}$ of labels. □

7.2 The invariant τ

Definition 7.2. Let M be a graded $\mathbb{F}[U]$-module that is finite-dimensional in each degree. Record degrees by *level*, meaning half the grading, so that U lowers the level by 1. Define $\tau(M)$ to be the multiset of levels k, where k appears with multiplicity $\dim(M_k/U \cdot M_{k+1})$.

τ is an isomorphism invariant, and shifting the grading translates τ. If $M = \mathcal{T}^+ \oplus \bigoplus_l \mathbb{F}[U]/U^{e_l}$, then $\tau(M)$ is the multiset of the top levels of the finite summands, because U maps onto $\mathcal{T}^+$; in particular $|\tau(M)|$ is the number of finite summands. Clearly $\tau(M \oplus M') = \tau(M) \cup \tau(M')$.

7.3 The level model and the structure of each class

Fix a class $c \in \mathbb{Z}/N$. Let $s_1 < s_2 < \cdots < s_m$ be the integers $s \equiv c \pmod N$ with $|s| \le g-1$. Consecutive ones differ by N, and

$$m = n_c := \#\{s \equiv c \pmod N : \ |s| \le g-1\}.$$

Level model. By (H2) we may compute with the homology-level maps U^{V_s} and U^{H_s} between copies of $\mathcal{T}^+$.

- Record each tower $\mathcal{T}^+$ by its *bottom level*.
- Regrade each B-tower so that the cone maps preserve levels. This is possible: in one class the A's and B's form a connected graph (each B is joined to its at most two adjacent A's), and the relative grading of the cone is homogeneous. A map U^e from a tower with bottom x then lands in a tower with bottom $x+e$.
- Under this regrading, an A-element at level k maps to the B-element at level k in each adjacent B that is *present at level* k, meaning that its bottom is $\le k$. All exponents are ≥ 0, so every present B has its adjacent A's present.
- So at level k the cone map is the coboundary $\delta_k \colon \mathbb{F}^{\text{vertices}} \to \mathbb{F}^{\text{edges}}$ of a graph G_k. Its vertices are the A's with bottom $\le k$, and its edges are the B's with bottom $\le k$, each B joining its adjacent A's.
- U acts on cochains by restriction from G_{k+1} to G_k.
- The cone has two terms and D commutes with U, so $H(\text{cone}) = \ker D \oplus \operatorname{coker} D$ as graded $\mathbb{F}[U]$-modules.

The true relative grading is 2·level+constant on each side, because both satisfy the same constraints on a connected graph. So τ, computed in levels, is the invariant τ above up to translation.

Proposition 7.3 (structure of a class)**.** *Let c and $s_1 < \cdots < s_m$ be as above.*

(A) Let $0 < p \le 2g-2$ and $m \ge 1$. Then $\tau(HF^+(S^3_p(K), c)) = \{y - 1 : \ y \in Y_c\}$, where Y_c is the multiset of the $m-1$ interior peak *levels*

$$y_j = \frac{\mu_j^2 - p^2/4}{2p}, \qquad \mu_j = s_j + \frac{p}{2} \qquad (1 \le j \le m-1).$$

In particular $|\tau| = m - 1$.

(B) Let $p = -N < 0$ and $m \geq 1$. Let Π_c be the multiset of the $m+1$ peak levels

$$b(t) = -\frac{u^2 - N^2/4}{2N}, \qquad u = t + \frac{N}{2}, \qquad t \in \{s_1 - N,\, s_1,\, s_2, \ldots, s_m\},$$

and let M be its maximum. Then $\tau(HF^+(S^3_p(K), c))$ is $\{y - 1 : \ y \in \Pi_c\}$ with one copy of $M - 1$ removed. In particular $|\tau| = m$.

Neither multiset depends on K.

Proof of (A). In class c the A's and B's form one infinite path $\cdots A_{s-p} - B_s - A_s - B_{s+p} - \cdots$. Writing $x(s)$ for the bottom of A_s, the bottoms must satisfy $x(s+p) - x(s) = H_s - V_{s+p}$, since B_{s+p} receives U^{H_s} from A_s and $U^{V_{s+p}}$ from A_{s+p}. One solution is

$$x(s) = \frac{s(s-p)}{2p} - V_s;$$

indeed $x(s+p) - x(s) = s - V_{s+p} + V_s = H_s - V_{s+p}$, using $H_s = V_s + s$. The B between s and $s+p$ has bottom

$$y = x(s) + H_s = \frac{s(s+p)}{2p} = \frac{\mu^2 - p^2/4}{2p}, \qquad \mu = s + \frac{p}{2}.$$

We have $x(s) \to +\infty$ as $|s| \to \infty$: for $s \geq g$, $V_s = 0$, and for $s < 0$, $V_s = V_{-s} + |s| \leq g + |s|$. So the quadratic term dominates, and each G_k is a finite subforest of a path.

The coboundary of a finite forest is onto. So D is surjective and $HF^+ \cong \ker D$. At level k, $\ker \delta_k$ is the space of functions on the components of G_k, and U is restriction of functions. Hence $\dim(M_k/U \cdot M_{k+1})$ is the number of *merges*: the number of components of G_k minus the number of components of G_{k+1} that contain a vertex of G_k.

Edges with a zero exponent. $V_s = 0$ iff $s \geq g$, and $H_s = 0$ iff $s \leq -g$. A B with both exponents 0 would be B_{s+p} with $s \leq -g$ and $s + p \geq g$, so $p \geq 2g$, which is excluded. A B with exactly one exponent 0 appears at the same level as its endpoint A_e on that side. That endpoint is new at that level. Its other edge has a positive exponent at A_e, since $V_e = H_e = 0$ would need $e \geq g$ and $e \leq -g$; so that other edge appears strictly later. The far endpoint of B is old, since B's exponent there is positive. So at this level B attaches the new vertex A_e to one old component, and merges no old components.

Edges with both exponents ≥ 1. The edge B_{s+p} between A_s and A_{s+p} has exponents $H_s \geq 1$ (iff $s \geq -(g-1)$) and $V_{s+p} \geq 1$ (iff $s + p \leq g - 1$). So these are exactly the $m - 1$ edges between consecutive s_j, s_{j+1}: the interior peaks. Each appears strictly after both of its endpoints, and since G is a path, each one joins two distinct components of the old graph.

So the merges from level k to level $k+1$ are the interior peaks at level $k+1$. Hence $\tau = \{y - 1 : y \text{ an interior peak}\}$ and $|\tau| = m - 1$. □

Proof of (B). Here $h_s \colon A_s \to B_{s-N}$. Let C be the class-c cone.

Truncation, step 1. Let $P = \{A_s : s < g\} \cup \{B_t : t < g\}$. It is a subcomplex: for $s < g$, A_s maps to B_s and B_{s-N}, and both indices are $< g$. The quotient $Q = C/P = \{A_s, B_s : s \geq g\}$ has differential $D_Q = v_s + h_s$, where $v_s \colon A_s \to B_s$ is the identity since $V_s = 0$, and h_s lowers the index by N and is present only while $s - N \geq g$. On finitely supported chains, the identity plus a part that lowers the index, with indices bounded below, is bijective: invert it by a finite Neumann series. So Q is acyclic and $H(C) \cong H(P)$.

Truncation, step 2. Let $P' = \{A_s : |s| < g\} \cup \{B_t : -g - N < t < g\}$. It is a subcomplex of P: for $|s| < g$ we have $s - N > -g - N$. The quotient $R_0 = P/P' = \{A_s : s \leq -g\} \cup \{B_t : t \leq -g - N\}$

has differential $h_s + v_s$, where $h_s\colon A_s \to B_{s-N}$ is the identity since $H_s = 0$, and $v_s\colon A_s \to B_s$ is present only while $s \le -g - N$. Relative to the pairing $A_s \leftrightarrow B_{s-N}$, the second part raises the B-index by N, and indices are bounded above. So it is again identity plus nilpotent on finitely supported chains, and R_0 is acyclic. Hence $H(C) \cong H(P')$.

Shape of P'. The class-c integers in $(-g-N, g)$ are $s_1 - N, s_1, \dots, s_m$, because $s_1 - N \le -g < s_1$ and $s_m < g \le s_m + N$. So P' is the zigzag

$$B_{s_1-N} - A_{s_1} - B_{s_1} - A_{s_2} - \cdots - A_{s_m} - B_{s_m}.$$

Every exponent on it is ≥ 1: $V_{s_j} \ge 1$ since $s_j \le g-1$, and $H_{s_j} = V_{-s_j} \ge 1$ since $-s_j \le g-1$. The two end B's each have one A-neighbour. Treat them as edges to a *ground* vertex, at which cochains take the value 0. Then $G \cup \{\text{ground}\}$ is a single cycle with $m+1$ edges.

Levels. Put $x(s) = -\frac{s(s+N)}{2N} - V_s$ and $b(t) = -\frac{t(t+N)}{2N} = -\frac{u^2 - N^2/4}{2N}$, $u = t + \frac{N}{2}$. Then B_s has bottom $x(s) + V_s = b(s)$, and since $H_s = V_s + s$,

$$x(s) + H_s = b(s) + s = \frac{-s^2 - sN + 2sN}{2N} = -\frac{s(s-N)}{2N} = b(s-N),$$

as required for B_{s-N}.

The cokernel part. $\operatorname{coker}\delta_k = H^1(G_k \cup \text{ground}, \text{ground})$, which is $H^1(G_k \cup \text{ground})$ because the ground is one point. This is 0 until the cycle closes, at the maximal peak level M. From then on it is $\mathbb{F}$, and U (restriction of cochains) is an isomorphism. So the cokernel part is a single tower and contributes nothing to τ.

The kernel part. $\ker\delta_k$ is the space of functions on the components of G_k that do not contain the ground. A vertex that is new at level $k+1$ has its edges strictly higher, since all exponents are ≥ 1. So it is isolated at level $k+1$, and the new edges at level $k+1$ join only old vertices or the ground. Hence $\dim(\ker_k / U \cdot \ker_{k+1})$ is the number of merges among old components, including merges into the ground component. That number is $\#(\text{new edges at level } k+1) - \Delta b_1$, and b_1 increases (by 1) only at level M. So $\tau = \{y - 1 : y \in \Pi_c\}$ with one copy of $M-1$ removed, and $|\tau| = (m+1) - 1 = m$. □

7.4 Divisibility

Proposition 7.4. *Suppose $2 \le |p| \le 2g-2$ and $S^3_p(K) \cong L \# Y''$ with L an L-space of order $a \ge 2$. Then $|p|$ divides $2g-1$.*

Proof. Write $2g - 1 = QN + r$ with $0 \le r < N$. Then $Q \ge 1$, since $N \le 2g-2$. The integers in $[-(g-1), g-1]$ are $2g-1$ consecutive integers, so $n_i = Q+1$ on a set S of r cyclically consecutive residues, and $n_i = Q$ elsewhere. In particular every $n_i \ge 1$.

By Proposition 7.3, $|\tau(HF^+(S^3_p(K), i))|$ equals $n_i - 1$ if $p > 0$ and n_i if $p < 0$; either way it is a strictly increasing function of n_i. By Lemma 7.1 and the invariance of τ up to translation, $|\tau|$ is invariant under $i \mapsto i + b$ with $b \not\equiv 0$. So S is invariant under translation by b.

If $0 < r < N$, then S is a proper nonempty cyclic interval in $\mathbb{Z}/N$. It has exactly one start point (an $x \in S$ with $x - 1 \notin S$), and a translation preserving S must fix that start point. That is impossible for $b \not\equiv 0$. So $r = 0$, i.e. N divides $2g-1$. □

7.5 The key contradiction

Theorem 7.5. *Let K be a nontrivial positive L-space knot of genus g, and p a nonzero integer with $|p| \ge 2$, $|p|$ dividing $2g-1$ and $|p| < 2g-1$. Then $S^3_p(K)$ is not homeomorphic to $L \# Y''$ for any L-space L with $|H_1(L)| \ge 2$ and any closed oriented 3-manifold Y''.*

Proof. Let $N := |p|$ and $n := (2g-1)/N$. Both are odd, $N \geq 3$ and $n \geq 3$. In particular $|p| \leq (2g-1)/3 \leq 2g-2$, so Proposition 7.3 applies. Suppose such a splitting exists, and let b be as in Lemma 7.1.

(i) The comparison. Let t be the representative of b with $0 < |t| \leq (N-1)/2$; it exists because N is odd and $b \not\equiv 0$. By Lemma 7.1 and the invariance of τ:

$$(\star) \qquad \tau \text{ of class } 0 \text{ and } \tau \text{ of class } t \text{ agree up to a translation.}$$

(ii) Arithmetic. For $|c| \leq (N-1)/2$, the integers $s \equiv c$ with $|s| \leq g-1$ are exactly $c+kN$ for $|k| \leq (n-1)/2$. Indeed $|c+kN| \leq (N-1)/2 + (n-1)N/2 = g-1$, while $|k| \geq (n+1)/2$ gives $|c+kN| \geq (nN+1)/2 > g-1$. So $m = n$ in every class.

Key fact. Let $u = c + (k+\frac{1}{2})N$ and $u' = c + (k'+\frac{1}{2})N$ with $k \neq k'$. If $u^2 = u'^2$, then $u = -u'$, so $2c + (k+k'+1)N = 0$. Hence $N \mid 2c$, so $N \mid c$ because N is odd, and $c = 0$. So for $c = t$, distinct k give distinct values of u^2, hence distinct levels in both formulas of Proposition 7.3.

(iii) Case $p > 0$. Class 0: $s_j = (j - \frac{n+1}{2})N$, so the interior peaks have $\mu_j = s_j + N/2 = (j - \frac{n}{2})N$ for $1 \leq j \leq n-1$. These are $\pm N/2, \pm 3N/2, \ldots, \pm(n-2)N/2$: nonzero $\pm$ pairs. So every element of τ has even multiplicity, and τ is nonempty since $n-1 \geq 2$. *Class t:* $\mu_j = t + (j - \frac{n+1}{2} + \frac{1}{2})N$, and by (ii) the $n-1$ levels are distinct. A translate of a nonempty multiset in which every multiplicity is even never has all multiplicities equal to 1. This contradicts $(\star)$.

(iv) Case $p < 0$. Class 0: the peaks are $t' = kN$ for $k = -\frac{n+1}{2}, \ldots, \frac{n-1}{2}$, so $u = t' + N/2$ runs over $\pm N/2, \pm 3N/2, \ldots, \pm nN/2$, each value of $|u|$ exactly twice. The maximal level M comes from $|u| = N/2$. Removing one copy of $M-1$ leaves the levels for $|u| = 3N/2, \ldots, nN/2$ still doubled, and these exist since $n \geq 3$. So τ has a repeated element. *Class t:* by (ii) the $n+1$ peak levels are distinct, so the n elements of τ are distinct. This contradicts $(\star)$. □

8 Proof of the Main Theorem

Assume, as in Section 4, that K is a nontrivial positive L-space knot of genus g, $p \neq 0$, and $Y = S^3_p(K)$ has $k \geq 3$ prime summands.

1. By Lemma 5.1, $Y \cong A \# B$ with B an integer homology sphere, $B \neq S^3$.

2. By Lemma 5.2, $Y \cong L \# Y''$ with L a lens space, hence an L-space, of order $a \geq 2$. So $|p| \geq 2$.

3. K is fibred with fibre of genus g and one boundary component (Theorem E). Proposition 6.1 gives $|p| \leq g$. If $g = 1$, this contradicts $|p| \geq 2$. So $g \geq 2$, and $2 \leq |p| \leq g \leq 2g-2$.

4. By Proposition 7.4, applied to the splitting of step 2, $|p|$ divides $2g-1$. Also $|p| \leq g < 2g-1$.

5. By Theorem 7.5, the splitting of step 2 cannot exist. This is a contradiction.

So $S^3_r(K)$ has at most two prime summands, for every L-space knot K and every slope r. □

9 Remarks

- For a fibred knot, a reducible integral surgery $S^3_p(K)$ with an integer homology sphere summand $B \neq S^3$ requires $|p| \leq g$ (Proposition 6.1). This uses only fibredness.

- Granting Theorem D, the argument of Section 8 shows more: a *reducible* surgery on an L-space knot never has a prime summand that is an integer homology sphere other than S^3, since steps 2–5 use only the homology sphere summand and the lens space summand. (Irreducible surgeries can be homology spheres, e.g. $S^3_{+1}(T(2,3))$ is the Poincaré sphere.) This agrees with the known examples, cables at the cabling slope, whose summands are $L(a,b)$ and $S^3_{b/a}(J)$.

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